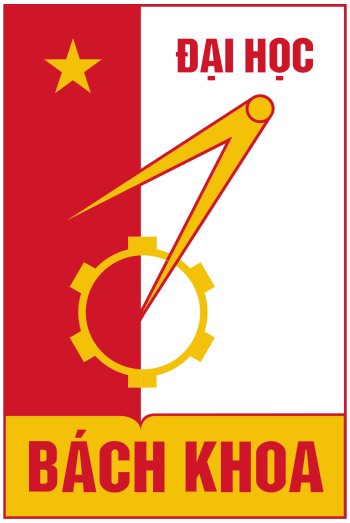
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Communication and Information Technology



KACoffee – Coffee Ordering and Chain Management System

Course: System Design - IT3120E

Class Code: 175881

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1 Introduction

1.1 Problem Definition

A coffee chain receives orders through several channels: its website, the counter of each branch and the telephone. Each order must reach the correct branch, retain the customer's drink choices and be updated as staff prepare it. Delivery orders also require a clear handover between store staff and the assigned shipper, so that responsibility for the order is clear at each stage.

When order records, customer information and promotion rules are managed separately, staff may need to enter the same information more than once. This creates opportunities for incorrect prices, missed order updates and inconsistent loyalty points. For example, points may be credited twice for one order, or not at all after a delivery is completed. Branch managers also need a shared view of completed sales and outstanding deliveries to coordinate daily operations, but such a view is difficult to build when each channel keeps its own records.

These difficulties increase with the number of branches and channels, because policies, order updates and sales figures must stay consistent across all of them.

The KACoffee project addresses these problems by bringing these activities into one system. Its initial focus is the transaction flow from account registration and order creation to preparation, collection or delivery, and completion.

1.2 Proposed Solution

To address the issues above, KACoffee is proposed as a web application that connects customer ordering with counter sales, delivery and branch management. The system will offer:

- **Public website:** Visitors can browse the public menu and register as a customer with a phone number.
- **Customer area:** Signed-in customers can place online orders for store pickup or delivery, review their orders and view their membership rank, points and notifications.
- **Staff interface:** Staff can create staff-assisted orders for walk-in customers and telephone callers, look up members by phone number, update preparation status and process bottle exchanges within their branch.
- **Shipper interface:** Shippers can view the deliveries assigned to them and record pickup, delivery progress and a successful or failed outcome.
- **Administration area:** Branch managers supervise orders, deliveries, reports and permitted employee records within their branch, while administrators manage system-wide products, branches, employee accounts, discount codes, settings and reports.

Each authenticated user will have an individual account and access appropriate to their role. A central database will keep product information, orders, customer membership records

and promotion rules consistent across branches. The system will also keep records of important business actions for traceability.

The project objectives are to:

- Provide a consistent ordering process across online, counter and telephone channels.
- Make responsibility and status visible throughout order preparation and delivery.
- Apply membership and discount rules consistently while protecting access to customer and branch data.
- Deliver an integrated application, supporting design documents and a demonstration of the five selected use cases.

2 Planning

2.1 Motivation

Coffee chains increasingly serve customers through more than one channel. A customer may order online for delivery in the morning, buy a drink at the counter in the afternoon and place an order by telephone in the evening. Each of these purchases should count towards the same membership, follow the same pricing and promotion rules and be visible to the branch that prepares it. When these channels are handled by separate tools or manual notes, staff re-enter the same information, managers lack a combined view of sales and deliveries, and loyalty points are credited inconsistently.

Our team is motivated to design a system in which every order, regardless of its source, follows one lifecycle from creation to completion, with each user group working in its own area. The project also gives the team a realistic setting in which to practise requirement analysis and system design before implementation: clear roles, branch-level access rules, state transitions for orders and deliveries, and business rules that must remain consistent across several use cases.

2.2 Initial System Request

2.2.1 Business Need

KACoffee will support coffee chain operations through shared order and membership records. Customers will be able to place orders and review their own information. Staff will record assisted purchases and prepare orders, shippers will update assigned deliveries, and management will supervise branch activity. The system must preserve the relationship between each transaction, the responsible branch and the users involved.

2.2.2 Business Requirements

The proposed system must fulfil the following key requirements:

- **Manage user accounts for different roles:** Customer, Staff, Shipper, Manager and Admin users each have their own login credentials, and Guest describes a visitor who has not signed in (Table 1). Staff, Shipper and Admin have role-specific sign-in pages and work areas, supported by a shared employee authentication service.
- **Register and authenticate users:** A customer registers with a unique phone number, full name and password; email is optional. A new account starts at the New membership rank with zero points. Customers sign in with their phone number and employees sign in with an assigned username. Passwords are stored as hashes, and authenticated sessions use access and refresh tokens. Inactive accounts and invalid credentials are rejected, and signed-in users can log out.
- **Provision internal accounts:** The initial Admin account is provisioned during system setup. Admin can create employee accounts and assign their roles and branches; a Manager can create and manage Staff and Shipper accounts within the permitted branch scope. Internal users do not use the public customer registration form.
- **Support online ordering:** A signed-in customer chooses active products available online, sets supported size, ice and sugar options, manages the cart, selects a branch and either store pickup or delivery (providing recipient and address information for delivery), and may apply a discount code. The system validates the code, calculates the final amount including any delivery fee and records the order with its items and initial status.
- **Support staff-assisted ordering:** Staff create orders for walk-in customers and telephone callers, with product selection, drink customisation, optional member lookup by phone number and eligible discounts. Customers without an account can also be served. The order belongs to the employee's branch and records the employee who created it, and authorised staff update its preparation and completion states.
- **Manage discount codes:** Admin views and creates discount codes using a percentage reduction or a fixed amount in VND, with an optional minimum order value, eligible membership ranks, an active state and optional validity dates. Order processing checks these conditions before applying a code and records its use with the transaction.
- **Track deliveries:** An assigned shipper updates progress through pickup, delivery in progress and a successful or failed outcome, and the system retains a status history. A successful delivery completes the related order, marks its offline payment as paid and triggers eligible loyalty points; a failed delivery remains available for management follow-up and reassignment.
- **Provide supporting functions:** Product and category management, branch and employee management, customer profiles, loyalty notifications, bottle exchanges, con-

figurable delivery fees, revenue and performance reports, branch targets and an audit trail for important actions.

Table 1: User groups and access rights

User group	**Account access**	**Main permissions**
Guest	No account required for public pages	Browse the public menu and register as a Customer.
Customer	Self-registration; phone number and password	Place online orders; view own profile, orders, membership points and notifications.
Staff	Provisioned account; username and password	Create assisted orders, look up members, update store orders and process bottle exchanges within the assigned branch.
Shipper	Provisioned account; username and password	View deliveries assigned to that shipper and record delivery progress or failure.
Manager	Provisioned account; username and password	Supervise orders, deliveries, reports and permitted employee records within the assigned branch.
Admin	Provisioned account; username and password	Manage system-wide products, branches, employee accounts, discounts, settings and reports.

The following business rules apply across the requirements above:

- **Order and payment rules:**
  + Orders identify their source as online, in-store or phone. New orders start as preparing and unpaid. Staff mark prepared orders as ready; authorised completion or successful delivery completes the transaction.
  + The initial release supports offline payment at the store or on receipt. When a transaction is completed, the system records it as paid. Revenue reports use completed orders.
  + Delivery is initially restricted to eligible addresses in Hanoi. The default fee is VND 20,000 and is configurable. Store pickup does not incur the delivery fee.
- **Membership and promotion rules:**
  + An eligible completed order earns one point for every VND 1,000 of qualifying item value after discounts. Delivery fees and free items are excluded, and points are credited once per order.
  + Membership thresholds are New at 0 points, Silver at 100, Gold at 300 and Diamond at 700. A discount restricted to membership ranks applies only to the ranks explicitly configured for that code.

+ Discounts must satisfy active status, any validity period, minimum order value and membership eligibility. The reduction does not exceed the item subtotal. Usage records support transaction history; a one-use-per-customer restriction is outside the initial rules.
+ Bottle exchanges accept positive multiples of five returned bottles. Each group of five earns one eligible bottled product. Staff can process an exchange for a member or an unregistered customer.

2.2.3 Business Value

The system brings value to the coffee chain and its users by:

- **Reducing repeated data entry:** Order and customer information is shared across channels and branches instead of being re-entered.
- **Applying policies consistently:** Common pricing, discount and membership checks help staff apply the same rules to every order.
- **Improving the customer experience:** Customers have a convenient ordering channel and visibility of their membership benefits.
- **Supporting branch supervision:** Managers can review completed sales and delivery outcomes by branch, while recorded actions help trace operational issues.

2.2.4 Special Issues or Constraints

- **Scope:** The initial release is a browser-based application with role and branch access controls. External payment gateways, SMS verification, live GPS tracking and ingredient-level stock management are outside this release.
- **Security and access control:** Protected screens and backend operations must check access rights. Customers access only their own records, shippers update only their own assigned deliveries, and branch restrictions apply to Staff and Manager operations.
- **Data consistency:** Item choices and totals must be preserved across the order lifecycle, and order and delivery states must remain consistent with each other.
- **System usability:** Interfaces should provide clear validation messages so that customers and staff can correct invalid input.
- **Schedule:** The team will prioritise the five core use cases when making schedule trade-offs.

2.3 Feasibility Analysis

2.3.1 Technical Feasibility

- **Current technology availability:** The proposed architecture separates the browser interface, the application API and the relational database. This division lets members work on their assigned flows while sharing API contracts and a common data model. The selected stack (Table 2) is suitable for the planned web features.
- **Developer expertise:**
  + **Familiarity with the application:** The business domain of ordering, counter sales and delivery has been organised into five core use cases, each with a named owner.
  + **Familiarity with the technology:** The main challenge is integrating authentication, pricing and order state changes, so these parts will be integrated early.
- **Infrastructure needs:** Docker Compose provides a consistent setup of the application and database for integration and demonstration. Local execution provides the baseline demonstration environment, so no dedicated server is required for the course project.
- **Project size and schedule:** The working estimate is eight weeks for a five-member team. This is a planning assumption to be aligned with the course timetable. Each member owns a core use case, while account handling, the data model and shared API contracts are coordinated across the team. A connected order flow will be integrated before the final phase so that defects are found while there is time to resolve them.
- **Technical risks:** The main risks are inconsistent API expectations, incorrect access checks and conflicting updates to orders or points. The team will agree request and response formats during design, centralise pricing and membership logic, and review state transitions before connecting the interfaces. Table 3 lists the initial risks and the planned responses.

Table 2: Proposed technology stack

Component	Proposed technology	Purpose
Frontend	React 19, TypeScript and Vite	Role-specific pages, reusable forms and navigation.
Client data	TanStack Query, Zustand and Axios	API data fetching, local interface state and HTTP requests.
Backend	Python and FastAPI	REST endpoints, request validation and business rules.
Data layer	MySQL 8.4, SQLAlchemy and Alembic	Relational storage, object mapping and schema migrations.
Identity	Password hashing and JWT tokens	User identity, sessions and role-based access checks.
Runtime	Docker Compose	A consistent application and database setup for integration and demonstration.

Table 3: Initial risks and mitigation

Risk	Planned response	Lead
Features exceed the available time	Prioritise UC001 to UC005; review added scope against the remaining schedule.	Trọng
Modules use inconsistent data or API formats	Agree shared contracts and migrations early; integrate small changes regularly.	Trọng with all members
Prices, discounts or points are calculated incorrectly	Keep calculations in shared backend services and check boundary-value scenarios.	Linh and Mạnh
A user accesses another role or branch's records	Enforce backend access checks and review both permitted and rejected requests.	Trọng with feature owners
Order and delivery states become inconsistent	Define valid transitions, retain status history and check success and failure flows.	Thành and Nhân
A member is delayed or unavailable	Record progress and handover notes; reassign a bounded task during the weekly review.	Trọng

Feasibility decision: The team proposes proceeding with the scoped academic project, subject to confirming the course schedule and the shared design. The first checkpoint requires working role-based sign-in and an agreed data model; the next checkpoint requires an integrated order flow. Any additional feature will be considered against these commitments and the time reserved for integration and documentation.

2.3.2 Operational Feasibility

- **Fit with business workflows:** The proposed workflows follow the responsibilities of a coffee business. Customers place their own orders, staff prepare orders and handle assisted purchases, and shippers act on assigned deliveries. Managers coordinate activity within their branches. Separate work areas keep the actions relevant to each user group visible.
- **Training and walkthroughs:** The team will prepare sample accounts for each role, sample products and a short usage guide. Operational review will cover a successful order as well as duplicate registration, invalid discounts and delivery failure.
- **Continuous improvement:** Feedback from these walkthroughs will guide wording and navigation changes before the final demonstration.

2.3.3 Economic Feasibility

- **Cost of development:** The course project relies on the team's existing computers and a local demonstration environment. The main resource demands are development time, integration work and documentation.
- **Optional costs:** Hosting, a domain and managed database services are optional costs if remote deployment is later required. The team will confirm service terms and obtain current quotations before making spending commitments.
- **Expected benefits:** The initial economic benefit will be demonstrated through fewer repeated entries, consistent calculations and easier access to operating information. Financial savings and commercial payback require real transaction volumes and operating costs before they can be estimated credibly.

2.3.4 Legal & Regulatory Feasibility

- **Data privacy:** KACoffee handles customer contact details, delivery addresses and employee account information. The team will minimise the information collected, restrict its visibility by role and use fictional records for development and demonstrations. Passwords and internal account data will not appear in public responses or reports.
- **Compliance:** Before any real deployment, the team will review applicable data protection obligations, third-party licences and service terms.

2.3.5 Market Feasibility

- **Target and value proposition:** The project targets a coffee chain that needs shared membership policies and coordinated sales across several branches. Its value proposition is the combination of customer ordering, assisted sales and internal delivery in

one application.

- **User fit:** The team will assess user fit through scenario walkthroughs and feedback on the proposed screens. Adoption levels and competitive advantage remain hypotheses until evaluated with actual operators and customers.

2.4 Initial Project Plan

The project will follow an **iterative and incremental approach**, with short development iterations within the proposed eight-week plan. Each iteration will have a defined set of tasks and a reviewable result. Early work will establish accounts, products, branches and shared data contracts; later iterations will connect ordering, promotions and delivery.

Key reasons for choosing this approach are:

- **Early integration:** Authentication, pricing and order state changes are shared by several use cases, so connecting them in early iterations exposes inconsistencies while there is still time to fix them.
- **Reviewable increments:** Each iteration produces a demonstrable result that can be reviewed by the team at the weekly meeting.
- **Clear ownership:** Each member owns one core use case end to end, from requirement analysis to documentation, which keeps responsibilities visible.
- **Risk reduction:** Testing and documentation accompany development rather than being left entirely to the final week.

2.4.1 Project Scope Estimation

Scope Size:

The project focuses on the transaction flow from account registration and order creation to preparation, collection or delivery, and completion. Development is organised around five selected use cases, with authentication, product data and branch data as the shared foundation. Each use case is listed with its acceptance focus:

- **UC001 Register as Customer:** Valid registration creates one account; invalid or duplicate input produces an understandable error without creating another account.
- **UC002 Place Online Order:** The order retains the chosen options, branch and fulfilment method, and its total reflects valid discounts and any delivery fee.
- **UC003 Create Staff-Assisted Order:** Both member and non-member purchases can be recorded, while invalid member information, unavailable products and ineligible discounts are handled clearly.
- **UC004 Manage Discount Codes:** A valid code reduces the item subtotal correctly; an inactive, expired or otherwise ineligible code is rejected.

- **UC005 Update Delivery Status:** Only the assigned shipper can update a delivery, invalid transitions are rejected, and the delivery outcome stays consistent with the related order.

Supporting functions (product, branch and employee management, bottle exchanges, delivery fees, reports, branch targets and the audit trail) are included to complete the order lifecycle, while external payment gateways, SMS verification, live GPS tracking and ingredient-level stock management are outside this release.

Time Estimate:

The project will span eight weeks, organised as short iterations and expressed in relative project weeks (Table 4). The team will confirm calendar dates against the course deadline and adjust task sizes while protecting time for integration and final review.

Table 4: Proposed schedule and milestones

Period	Planned work	Expected output
Week 1	Confirm scope, accounts, business rules and member responsibilities.	Agreed requirements, initial backlog and use case descriptions.
Week 2	Design the data model, screens and API contracts; establish the shared environment.	Schema, interface sketches, contracts and a running project skeleton.
Weeks 3 and 4	Implement registration, role-based sign-in, product access, online and assisted ordering.	Demonstrable UC001 to UC003 with basic access checks.
Weeks 5 and 6	Connect discount management, delivery and completion; integrate supporting functions.	UC004 and UC005 plus a connected order and loyalty flow.
Week 7	Review permissions and business rules, exercise failure cases and fix integration defects.	Reviewed test results, corrected defects and a stable demo dataset.
Week 8	Complete the report, usage guide and presentation; rehearse the demonstration.	Final application package, documentation and team demonstration.

2.4.2 Staffing Plan

The five members will own the five selected use cases. Ownership covers requirement analysis, interface and API design, implementation, tests and documentation for the assigned flow. Supporting functions are grouped with the nearest business area (Table 5).

Table 5: Team structure and responsibilities

Member	Role / use case	Planned responsibilities
Trọng	Team coordinator; Guest and UC001	Coordinate scope and progress; build registration and shared authentication; maintain role checks; coordinate integration and the final report.
Linh	Customer flow owner; UC002	Build menu selection, cart, online checkout and customer order views; coordinate product options, totals and customer membership information.
Thành	Staff flow owner; UC003	Build assisted ordering, member lookup and store order handling; support preparation updates and bottle exchange.
Mạnh	Administration owner; UC004	Build discount administration and eligibility rules; coordinate products, branches, employee management and operational reports.
Nhân	Delivery flow owner; UC005	Build assigned delivery views and status updates; coordinate assignment and reassignment with administration, order completion and delivery history.

Shared work will be coordinated as follows:

- **Shared design:** All members agree the core data model and review changes to shared entities. Trọng coordinates account identity and access rules. Linh and Thành agree the common order structure and item options, Mạnh provides discount validation rules, and Nhân agrees delivery outcomes with both order owners.
- **Peer review:** Linh reviews registration, Thành reviews online ordering, Mạnh reviews assisted ordering, Nhân reviews discount handling, and Trọng reviews delivery. The owner remains responsible for resolving review findings and keeping the related documentation current.
- **Communication:** The team holds one progress meeting each week and uses a shared group chat for blockers. Trọng summarises decisions, dependencies and any reassignment so that changes remain visible to everyone.
- **Change management:** Members work on separate Git branches and submit changes for peer review before merging. A feature is ready for integration when its main flow, validation behaviour, access rules and documentation have been reviewed. Changes to the database or shared APIs include an update to the agreed design, and new requirements are assessed for their impact on the five core use cases and the remaining schedule.
- **Quality review:** The planned review covers successful transactions, duplicate registration, invalid credentials, cross-account and cross-branch access, ineligible discounts, non-member counter orders and failed deliveries, and checks that completion awards points once and excludes delivery fees.

- **Final deliverables:** The application source code, database migrations and demonstration data, setup instructions, use case and design documentation, the test summary, user guidance and presentation slides. Each member will demonstrate the assigned flow and explain its connection to the shared system.

2.4.3 CASE Tools

The project makes use of the following CASE (Computer-Aided Software Engineering) and supporting tools:

- **UML diagram creation:** draw.io (diagrams.net) is used to draw the use case diagram and the swimlane activity diagrams; later design diagrams will use the same tool. Diagram sources are kept with the report so that they can be edited.
- **Database design:** Alembic migrations keep the relational schema under version control; a database client such as MySQL Workbench is planned for inspecting and documenting the schema.
- **API documentation:** The FastAPI interactive documentation is used to agree and check request and response formats.
- **Project management:** A GitHub task board records each task's owner, acceptance criteria, status and target week; Git branches and pull requests support peer review.
- **Documentation:** The report is written in LaTeX so that sections, tables and figures are numbered and cross-referenced consistently.

3 Analysis

3.1 General Use Case Diagram

Figure 1 depicts the general use case diagram of the KACoffee system. The system boundary contains the browser interface, the application API and the data managed by the system; the database, services and authentication tokens are internal parts of the system and are therefore not actors. External services such as payment gateways, SMS verification and GPS tracking are outside this release and do not appear in the diagram.

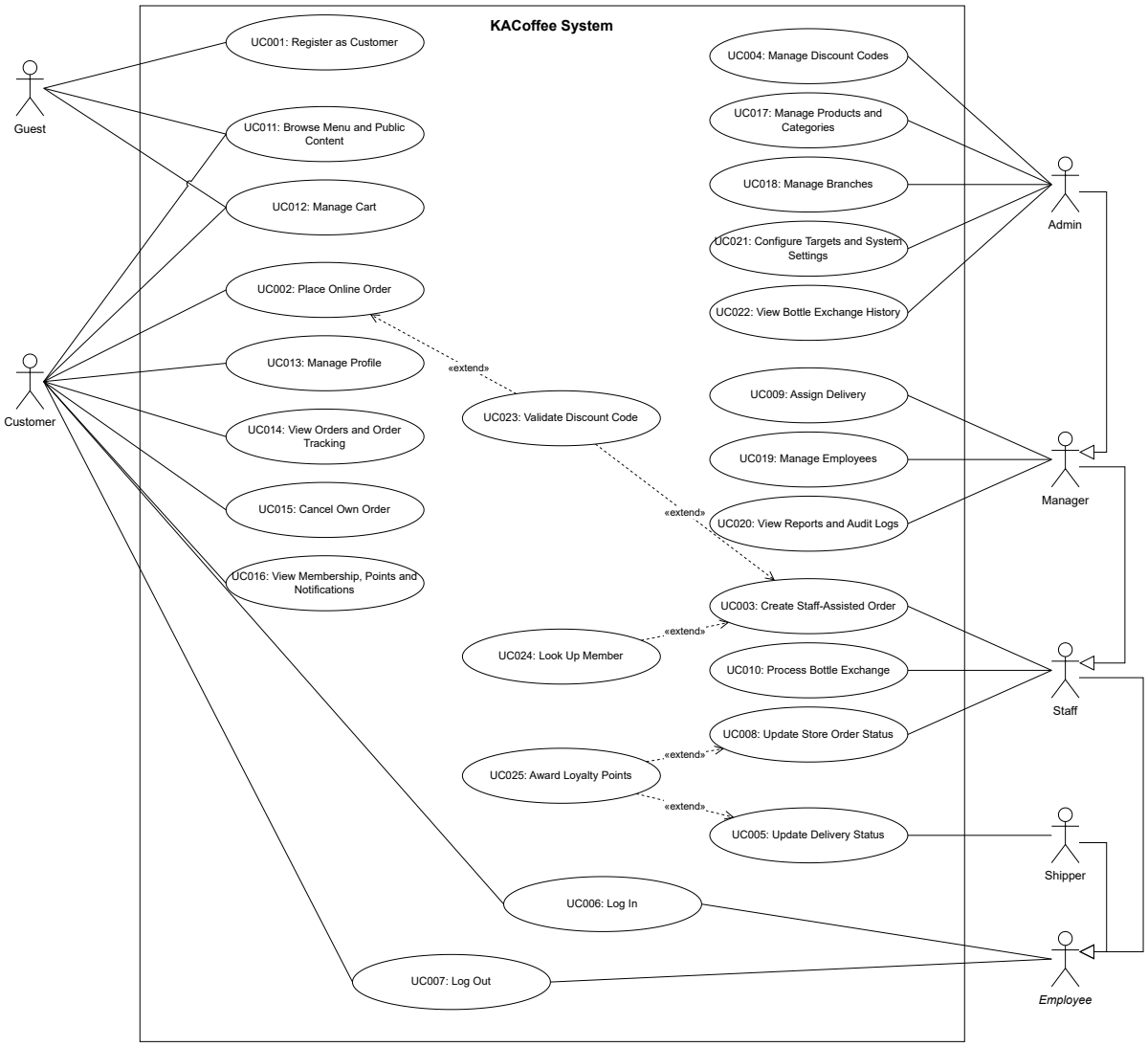


Figure 1: General use case diagram of the KACoffee system

The system has six actors:

- **Guest:** Represents a visitor who has not signed in. A Guest can browse the public menu and content, prepare a cart and register as a Customer. Upon successful registration, the visitor becomes a Customer.
- **Customer:** Represents a registered member who signs in with a phone number. Customers place online orders for pickup or delivery, manage their profile, review and cancel their own orders while they are still being prepared, and view their membership rank, points and notifications.
- **Staff:** Represents a store employee working at the counter of one branch. Staff create staff-assisted orders for walk-in customers and telephone callers, look up members by phone number, update the status of the branch's orders and process bottle exchanges.
- **Shipper:** Represents a delivery employee of a branch. Shippers view the deliveries assigned to them and record pickup, delivery progress and a successful or failed outcome.

- **Manager:** Represents the supervisor of one branch. Managers assign and reassign shippers, manage Staff and Shipper accounts of their branch and view the branch's reports, target progress and audit log. Managers may also use the Staff functions.
- **Admin:** Represents the system administrator. Admin manages system-wide products, categories, branches, employee accounts, discount codes, performance targets and system settings, and can view reports for all branches. Admin may also use the Staff and Manager functions.

To keep the diagram readable, actor generalisation is used where one role can perform every use case of another: Manager specialises Staff and Admin specialises Manager, so the Staff use cases are drawn only once. *Employee* is an abstract actor that groups the internal accounts (Staff and Shipper, and through them Manager and Admin) which sign in through the shared employee authentication service; it is not a separate user group.

Table 6 lists the use cases shown in the diagram. UC001 to UC005 are the five core use cases of the project; UC006 to UC010 are supporting use cases that complete the order lifecycle. These ten use cases are specified in Section 3.2 and modelled in Section 3.3. The remaining use cases are included to show the full scope of the system.

Table 6: Use case catalogue

Code	Use case	Actors
UC001	Register as Customer	Guest
UC002	Place Online Order	Customer
UC003	Create Staff-Assisted Order	Staff (also Manager, Admin)
UC004	Manage Discount Codes	Admin
UC005	Update Delivery Status	Shipper
UC006	Log In	Customer, Staff, Shipper, Manager, Admin
UC007	Log Out	Customer, Staff, Shipper, Manager, Admin
UC008	Update Store Order Status	Staff (also Manager, Admin)
UC009	Assign Delivery	Manager, Admin
UC010	Process Bottle Exchange	Staff (also Manager, Admin)
UC011	Browse Menu and Public Content	Guest, Customer
UC012	Manage Cart	Guest, Customer
UC013	Manage Profile	Customer
UC014	View Orders and Order Tracking	Customer

Code	Use case	Actors
UC015	Cancel Own Order	Customer
UC016	View Membership, Points and Notifications	Customer
UC017	Manage Products and Categories	Admin
UC018	Manage Branches	Admin
UC019	Manage Employees	Manager (own branch), Admin
UC020	View Reports and Audit Logs	Manager (own branch), Admin
UC021	Configure Targets and System Settings	Admin
UC022	View Bottle Exchange History	Admin
UC023	Validate Discount Code	— (extends UC002, UC003)
UC024	Look Up Member	— (extends UC003)
UC025	Award Loyalty Points	— (extends UC005, UC008)

UC023 to UC025 have no actor of their own; they are reached only through the base use cases they extend. The diagram uses the following «extend» relationships:

- **UC023 Validate Discount Code** «extend» UC002 and UC003: executed only when a discount code is entered. The same validation rules (active state, validity dates, minimum order value and eligible ranks) are shared by both ordering channels.
- **UC024 Look Up Member** «extend» UC003: executed only when the customer is a member and Staff enters a phone number.
- **UC025 Award Loyalty Points** «extend» UC005 and UC008: executed when an order linked to a member reaches the completed status, either by a successful delivery (UC005) or by completing a pickup order (UC008). Points are not awarded when an order is created in UC002 or UC003.

Each extension is inserted at the named extension point of the base use case, and the arrow points from the extending use case to the base use case. Signing in (UC006) is a precondition of the protected use cases rather than an included use case, and a failed delivery is described as an alternative flow of UC005 instead of a separate use case.

3.2 Use Case Details

Each use case is specified with the fields of the template, extended with Alternative Flows and Business Rules rows. The main event flow lists who performs each step; each alternative flow gives the step where it starts, its condition, the handling and where the flow continues

or ends. Flows labelled A are valid alternatives and flows labelled E are exceptions.

3.2.1 Guest's and Common Use Cases

UC001: Register as Customer

Use Case Code	UC001	**Use Case Name**	Register as Customer
Actor	Guest		
Usage	Allow a visitor without an account to create a customer membership account with a phone number, so that the visitor can place online orders and earn loyalty points.		
Activation Event	The Guest selects "Register" and submits the registration form.		
Preconditions	The Guest is not signed in and is on the KACoffee website.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Guest	Select "Register".
	2	System	Present the registration form: full name, phone number, password and an optional email.
	3	Guest	Fill out the form and submit it.
	4	System	Validate the required fields and their length constraints.
	5	System	Check that the phone number has not already been registered.
	6	System	Create the customer account at the New membership rank with zero points and add a welcome notification.
	7	System	Issue access and refresh tokens to start a signed-in session.
	8	System	Load the account summary and open the member area.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	A1 (3)	The Guest leaves the form.	The use case ends and no account is created.

	E1 (4)	Data is missing or invalid.	The system shows a validation message for each invalid field; the Guest corrects the data and continues from step 3.
	E2 (5)	The phone number is already registered.	The system shows “Phone already registered” and creates no account. The Guest enters another phone number (step 3) or signs in instead (UC006).
Postconditions	Success: exactly one new customer account exists at the New rank with zero points, a welcome notification is stored and the customer is signed in. Failure: no account is created.		
Business Rules	The phone number is unique and is the customer’s login identifier. Full name is 1–100 characters, phone number 8–20 characters and password 6–100 characters; email is optional. Passwords are stored as hashes. This release has no OTP or email verification step.		

UC006: Log In

Use Case Code	UC006	**Use Case Name**	Log In
Actor	Customer, Staff, Shipper, Manager, Admin		
Usage	Authenticate a registered user so that the user can access the work area of their role.		
Activation Event	The user selects “Log In” on the customer sign-in page or on a role-specific employee sign-in page (Staff, Shipper or Admin).		
Preconditions	The user has an active account: a customer account created through UC001, or an employee account provisioned by Admin or Manager.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	User	Open the sign-in page for their user group.
	2	System	Present the login form: phone number and password for customers, username and password for employees.
	3	User	Enter the credentials and submit the form.
	4	System	Verify that an active account matches the credentials.
	5	System	Issue access and refresh tokens and load the user’s profile and role.

	6	System	Open the work area of the role: member area for Customer, staff area for Staff, shipper area for Shipper and administration area for Manager and Admin.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	E1 (4)	The credentials are invalid or the account is inactive.	The system shows "Invalid credentials" without revealing which part is wrong; the user tries again from step 3.
Postconditions	The user is authenticated, the refresh token is recorded and the work area matching the user's role is displayed.		
Business Rules	Customers sign in with their phone number; employees sign in with an assigned username through a shared employee authentication service. Inactive accounts are rejected.		

UC007: Log Out

Use Case Code	UC007	**Use Case Name**	Log Out
Actor	Customer, Staff, Shipper, Manager, Admin		
Usage	End the user's current session on the browser.		
Activation Event	The user selects "Log Out".		
Preconditions	The user is signed in.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	User	Select "Log Out".
	2	System	Remove the stored tokens and profile from the browser.
	3	System	Show the signed-out header; a protected page redirects to the sign-in page of its portal.
Postconditions	The session is ended on the browser; protected screens require signing in again (UC006).		
Business Rules	The backend also offers a logout endpoint that revokes the refresh token so that it can no longer obtain new access tokens; the current web interface ends the session locally and does not call it.		

3.2.2 Customer's Use Cases

UC002: Place Online Order

Use Case Code	UC002	**Use Case Name**	Place Online Order
Actor	Customer		
Usage	Allow a signed-in customer to turn the prepared cart into an online order for store pickup or delivery, optionally applying a discount code.		
Activation Event	The Customer selects "Checkout" from the cart.		
Preconditions	The Customer is signed in. The cart contains at least one product with the chosen size, ice and sugar options.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Customer	Review the cart and select "Checkout".
	2	System	Present the checkout form with the active branches, the fulfilment methods and the order summary.
	3	Customer	Select a branch and a fulfilment method (store pickup or delivery) and enter the recipient's name and phone number.
	4	Customer	For delivery, enter the delivery address (address line, ward, district and city).
	5	Customer	Optionally enter a discount code and request a preview.
	6	System	Extension point *Validate Discount Code*: validate the code and show the discount amount in the updated summary.
	7	Customer	Confirm the order.
	8	System	Check the branch, the products (active and available online), the item options and, for delivery, the delivery city.
	9	System	Recalculate the item prices from product data, re-validate the discount code, if any, and add the delivery fee for delivery orders.

	10	System	Record the order (source online, status preparing, payment unpaid) with its items, the initial status history and the discount usage, if any.
	11	System	Clear the cart and display the order details with the order number.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	A1 (3)	The Customer chooses store pickup.	Step 4 is skipped and no delivery fee is charged.
	A2 (5)	The Customer has no discount code.	Steps 5–6 are skipped; the flow continues at step 7.
	E1 (6)	The code does not exist, is inactive, is not yet valid or has expired, the rank is not eligible, or the subtotal is below the minimum order value.	The system shows the reason and applies no discount. The Customer changes the code (step 5) or clears it and continues (step 7).
	E2 (8)	The cart is empty, or a product is inactive or not available online.	The order is rejected; the Customer updates the cart (step 1).
	E3 (8)	The branch is not available.	The order is rejected; the Customer selects another branch (step 3).
	E4 (8)	The delivery city is missing or not recognised as Hanoi.	The order is rejected; the Customer corrects the address or chooses store pickup (step 3).

	E5 (9)	The discount code kept in the form is no longer valid, or was kept after a failed preview.	The order is rejected and nothing is recorded; the Customer removes or changes the code (step 5).
Postconditions	Success: an online order exists with status preparing and payment unpaid; it keeps the chosen options, branch and fulfilment method, and its total equals the item subtotal minus the discount plus any delivery fee. No delivery record is created and no points are awarded at this stage. Failure: no order is recorded.		
Business Rules	Payment is offline (at the store or on receipt). The default delivery fee is VND 20,000 and is configurable; store pickup has no fee. Delivery is restricted to Hanoi; only the city name is checked and the other address fields are optional. The discount never exceeds the item subtotal. Each line has a quantity of 1–50.		

3.2.3 Staff's Use Cases

UC003: Create Staff-Assisted Order

Use Case Code	UC003	**Use Case Name**	Create Staff-Assisted Order
Actor	Staff (Manager and Admin may also use this function). Supporting participant: Store Customer / Caller, who does not use the system directly.		
Usage	Allow staff to record an order for a walk-in customer or a telephone caller, with or without a membership account.		
Activation Event	Staff receives an order request at the counter or by phone and opens the point-of-sale (POS) order screen.		
Preconditions	Staff is signed in and assigned to a branch. At least one product is available for in-store ordering.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Store Customer / Caller	Tell the staff member which drinks they want.
	2	Staff	Open the POS order screen and select the order source (in-store or phone).
	3	System	Display the products available for in-store ordering.

	4	Staff	Select products, quantities, size, ice and sugar levels and the serving option (dine-in or takeaway).
	5	System	Update the line items and the running subtotal.
	6	Staff	Ask whether the customer is a member; if so, enter the member's phone number and look it up.
	7	System	Extension point *Look Up Member*: display the member's name, membership rank and current points.
	8	Staff	Enter the recipient information, the fulfilment method and, if provided, a discount code.
	9	Staff	Confirm the order.
	10	System	Validate the products, the recipient and delivery data and the member; at the extension point *Validate Discount Code*, validate the code, if any; then calculate the official total.
	11	System	Record the order in the staff member's branch (source in-store or phone, status preparing, payment unpaid) with the creating employee, the items, the status history and the discount usage, if any.
	12	System	Display the created order with the server-calculated discount and total.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	A1 (2–9)	Staff cancels before confirming.	The use case ends and nothing is recorded.
	A2 (6)	The customer is not a member.	Staff leaves the phone number empty; step 7 is skipped and the order continues at step 8 without member benefits.

	E1 (7)	No member has this phone number.	The system reports that the member was not found. Staff corrects the number (step 6) or clears it to continue as a non-member (step 8); if the unknown number is left in the form, the order is rejected at step 10.
	A3 (8)	The customer has no discount code.	Staff leaves the code empty and continues at step 9.
	E2 (10)	A discount code is entered without an identified member, or the code is invalid, expired, below the minimum order value or not eligible for the rank.	The order is rejected and nothing is recorded; Staff removes or changes the code (step 8).
	E3 (10)	The delivery city is missing or not recognised as Hanoi, or a delivery order contains a dine-in item.	The order is rejected; Staff corrects the items (step 4) or the delivery data (step 8).
Postconditions	Success: an order with source in-store or phone, status preparing and payment unpaid belongs to the staff member's branch and records the employee who created it. Loyalty points are awarded only when the order is later completed (UC005 or UC008). Failure: no order is recorded.		
Business Rules	Source (in-store or phone), fulfilment method (pickup or delivery) and serving option (dine-in or takeaway) are independent attributes. A discount code requires an identified member. Unavailable products are not offered on the POS screen.		

UC008: Update Store Order Status

Use Case Code	UC008	**Use Case Name**	Update Store Order Status
Actor	Staff (Manager and Admin may also use this function)		

Usage	Move an order of the branch through preparation: mark it ready, complete a pickup order or cancel an order that is still being prepared.		
Activation Event	Staff selects an order on the order status screen and chooses a new status.		
Preconditions	Staff is signed in. The order belongs to the staff member's branch (Admin is not restricted to a branch).		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Staff	Open the order status screen.
	2	System	Display the branch's orders with their current status.
	3	Staff	Select a preparing order and choose "ready".
	4	System	Check that the order belongs to the branch and that the transition is allowed.
	5	System	Update the order status and record the status history and an audit log entry.
	6	System	Notify the member, if any; for a delivery order, create a delivery record awaiting shipper assignment.
	7	System	Display the updated order.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	A1 (3)	Staff cancels an order that is still preparing.	Steps 4–5 apply; the order and its payment status become cancelled and no notification is sent. The flow continues at step 7.
	A2 (3)	Staff completes a ready pickup order.	Steps 4–5 apply; the system records the payment as paid and, at the extension point *Award Loyalty Points*, credits the member's points once. The flow continues at step 7.

	E1 (4)	The transition is not allowed, e.g. completing a delivery order or changing a completed or cancelled order.	The system shows “Invalid status transition”; Staff chooses another action (step 3).
	E2 (4)	The order belongs to another branch.	Access is denied and the use case ends.
Postconditions	The order has a new status and a status history entry. A ready delivery order has a delivery record awaiting assignment (UC009); a completed pickup order is paid and, for a member, its points are credited.		
Business Rules	Allowed transitions: preparing → ready or cancelled; ready → completed for pickup orders only. Delivery orders are completed through UC005. Points are credited once per order and exclude delivery fees and free items.		

UC010: Process Bottle Exchange

Use Case Code	UC010	**Use Case Name**	Process Bottle Exchange
Actor	Staff (Manager and Admin may also use this function). Supporting participant: Store Customer.		
Usage	Record the exchange of returned bottles for eligible bottled products, for a member or an unregistered customer.		
Activation Event	A customer brings returned bottles to the counter and Staff opens the bottle exchange screen.		
Preconditions	Staff is signed in and assigned to a branch. At least one active bottled product exists.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Store Customer	Hand over the returned bottles.
	2	Staff	Open the bottle exchange screen.
	3	System	Display the bottled products eligible as rewards.
	4	Staff	Enter the number of returned bottles, choose the reward product and optionally enter the member’s phone number and a note.

	5	Staff	Submit the exchange.
	6	System	Check that the number of bottles is a positive multiple of five, that the reward product is an active bottled product and that the member exists if a phone number was entered.
	7	System	Calculate the reward quantity (one product for every five bottles) and record the exchange with the branch, the staff member and the member, if any.
	8	System	Display the recorded exchange with the reward quantity.
	9	Staff	Hand the reward products to the customer.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	E1 (6)	The number of bottles is not a positive multiple of five.	The system rejects the exchange; Staff corrects the quantity (step 4).
	E2 (6)	No member has the phone number entered.	The system reports that the member was not found; Staff corrects or clears the phone number (step 4).
	E3 (6)	The reward product is not an active bottled product.	The system rejects the exchange; Staff chooses another product (step 4).
Postconditions	An exchange record exists for the branch with the returned quantity, reward product and quantity, the processing employee and the member, if identified; an audit log entry is stored.		
Business Rules	Reward quantity = returned bottles / 5; for example 10 bottles give 2 products, while 7 bottles are rejected. Exchanges do not award loyalty points.		

3.2.4 Shipper's Use Cases

UC005: Update Delivery Status

Use Case Code	UC005	**Use Case Name**	Update Delivery Status
Actor	Shipper		
Usage	Allow the assigned shipper to record the progress of a delivery from pickup at the store to a successful or failed outcome, keeping the related order consistent.		
Activation Event	The Shipper opens the list of assigned deliveries and acts on one of them.		
Preconditions	The Shipper is signed in. To update a delivery, it must be assigned to this shipper (UC009) and be in a status that allows the action. The related order stays in the ready status while it is being delivered.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Shipper	Open the deliveries list.
	2	System	Display the deliveries assigned to this shipper with the address, item summary and current status.
	3	Shipper	Collect the order at the store and confirm pickup for an assigned delivery.
	4	System	Check the assignment and status, set the delivery to picked up and record the time and status history.
	5	Shipper	Confirm the start of the delivery.
	6	System	Set the delivery to delivering and record the time and status history.
	7	Shipper	Hand over the order, collect the offline payment and confirm successful delivery.
	8	System	Set the delivery to delivered, set the related order to completed and its payment to paid, and record the status history.
	9	System	Extension point *Award Loyalty Points*: if the order belongs to a member and no points have been awarded yet, credit the points and upgrade the membership rank when a threshold is reached.

	10	System	Display the updated delivery.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	A1 (2)	No assigned deliveries.	The system shows an empty list and the use case ends.
	A2 (3, 5)	Shortened path.	The system also accepts starting the delivery directly from assigned (step 5) and confirming success directly from picked up (step 7).
	A3 (3, 5 or 7)	The order cannot be delivered, e.g. the customer is absent or the address is wrong.	The Shipper enters a failure reason and reports a failed delivery. The system sets the delivery to failed, keeps the order in the ready status, records the reason in the delivery history and notifies the member, if any. The use case ends; reassignment is handled in UC009.
	E1 (4, 6, 8)	The delivery is not assigned to this shipper.	Access is denied and the use case ends.
	E2 (4, 6, 8)	The action is not allowed from the current status.	The system rejects the update; the delivery keeps its status and the Shipper chooses an allowed action.
	E3 (in A3)	No failure reason is entered.	The system asks for a reason (1–255 characters); the Shipper enters it and A3 continues.
Postconditions	Every change is stored in the delivery status history. Delivered: the order is completed and paid, and member points are credited once. Failed: the order remains ready and is available for follow-up and reassignment (UC009).		
Business Rules	Delivery lifecycle: pending assignment → assigned → picked up → delivering → delivered, or failed from assigned, picked up or delivering. Intermediate delivery states do not change the order status. Points = floor(eligible amount / 1,000), where the eligible amount is the value of non-free items after discount, excluding the delivery fee. Thresholds: Silver 100, Gold 300, Diamond 700.		

3.2.5 Manager's and Admin's Use Cases

UC009: Assign Delivery

Use Case Code	UC009	**Use Case Name**	Assign Delivery
Actor	Manager (within the assigned branch), Admin		
Usage	Assign a shipper of the same branch to a ready delivery order, or reassign the delivery to another shipper, for example after a failed attempt.		
Activation Event	The Manager opens the deliveries screen and chooses a shipper for a delivery.		
Preconditions	The Manager or Admin is signed in. The related order uses delivery and has status ready (UC008, or a failed delivery in UC005).		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Manager	Open the deliveries screen.
	2	System	Display the deliveries in scope (the Manager's branch, or all branches for Admin) with their status and shipper.
	3	Manager	Select a delivery and choose an active shipper.
	4	System	Check that the delivery is in the Manager's branch, that the order is a ready delivery order and that the selected employee is an active shipper of the same branch.
	5	System	Set the shipper and the status assigned, record the assignment time, the status history and an audit log entry (assigned or reassigned).
	6	System	Display the updated delivery; it now appears in the shipper's list (UC005).
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	E1 (4)	The delivery belongs to a branch other than the Manager's.	Access is denied and the use case ends.

	E2 (4)	The order is not a ready delivery order.	The system rejects the assignment and the use case ends.
	E3 (4)	The employee is not an active shipper or belongs to another branch.	The system rejects the assignment; the Manager chooses another shipper (step 3).
Postconditions	The delivery is assigned to one shipper of the order's branch, and the change is recorded in the delivery history and the audit log.		
Business Rules	A Manager acts only within the assigned branch. Reassignment uses the same checks and sets the delivery back to assigned.		

UC004: Manage Discount Codes

Use Case Code	UC004	**Use Case Name**	Manage Discount Codes
Actor	Admin		
Usage	Allow the administrator to review the existing discount codes and create a new code that customers and staff can apply to orders.		
Activation Event	The Admin opens the "Promotions" (discount codes) page.		
Preconditions	The Admin is signed in with the admin role. To restrict a code by rank, the membership ranks exist.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Open the discount codes page.
	2	System	Display the list of existing codes and the creation form with the available membership ranks.
	3	Admin	Enter the code, an optional description, the discount type (percentage or fixed VND amount), the value, an optional minimum order value and the eligible ranks, if any.
	4	Admin	Submit the form.
	5	System	Normalise the code (trimmed, upper case), validate the fields and check that the code is unique.

	6	System	Save the active discount code together with its rank eligibility links.
	7	System	Show a success message, refresh the list and reset the form.
Alternative Flows	**ID (step)**	**Condition**	**Handling and resume point**
	A1 (3)	No rank is selected.	The code is created without a rank restriction.
	A2 (3)	The Admin leaves the page.	The use case ends and no code is created.
	E1 (5)	The code already exists.	The system shows “Discount code already exists”; the Admin changes the code (step 3).
	E2 (5)	Values are invalid, e.g. a value below 1 or a code shorter than 3 characters.	The system shows validation messages; the Admin corrects the form (step 3).
Postconditions	Success: a new discount code exists with the active state and zero or more eligible ranks, and can be applied in UC002 and UC003. Failure: no code is created.		
Business Rules	Percentage discount = subtotal × value / 100; fixed discount = value, never exceeding the item subtotal. A code applies only if it is active, within its optional validity dates, the minimum order value is met and the member’s rank is eligible (no ranks configured means no rank restriction). Validity dates, the active flag and later edits are supported by the backend API but are not exposed on the current admin screen; the form always creates an active code.		

3.3 Activity Diagram

Each activity diagram uses swimlanes to show who performs each action. The left lane is the actor that starts the use case and the right lane is the KACoffee System; UC003 adds a “Store Customer / Caller” lane and UC010 a “Store Customer” lane. Decision nodes carry guard conditions in brackets, and error branches return to the step where the user corrects the input or end the activity.

3.3.1 Guest’s and Common Use Cases

UC001: Register as Customer

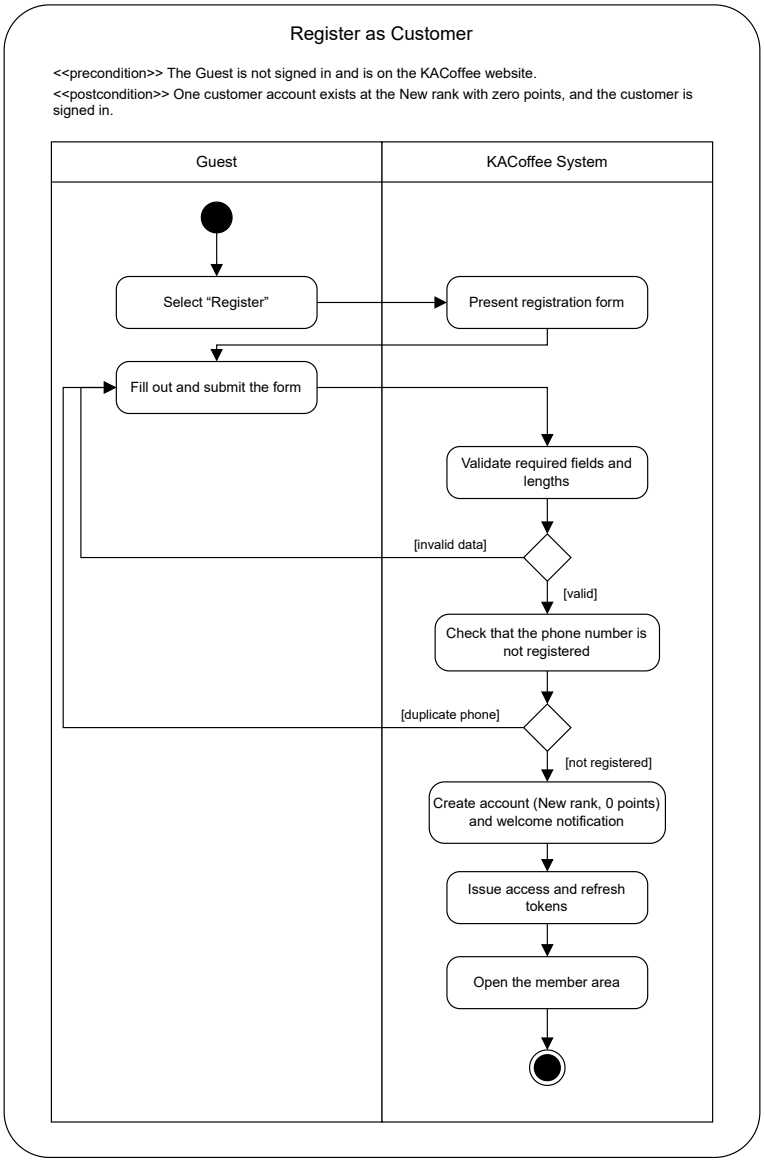


Figure 2: Activity diagram of UC001 Register as Customer

UC006: Log In

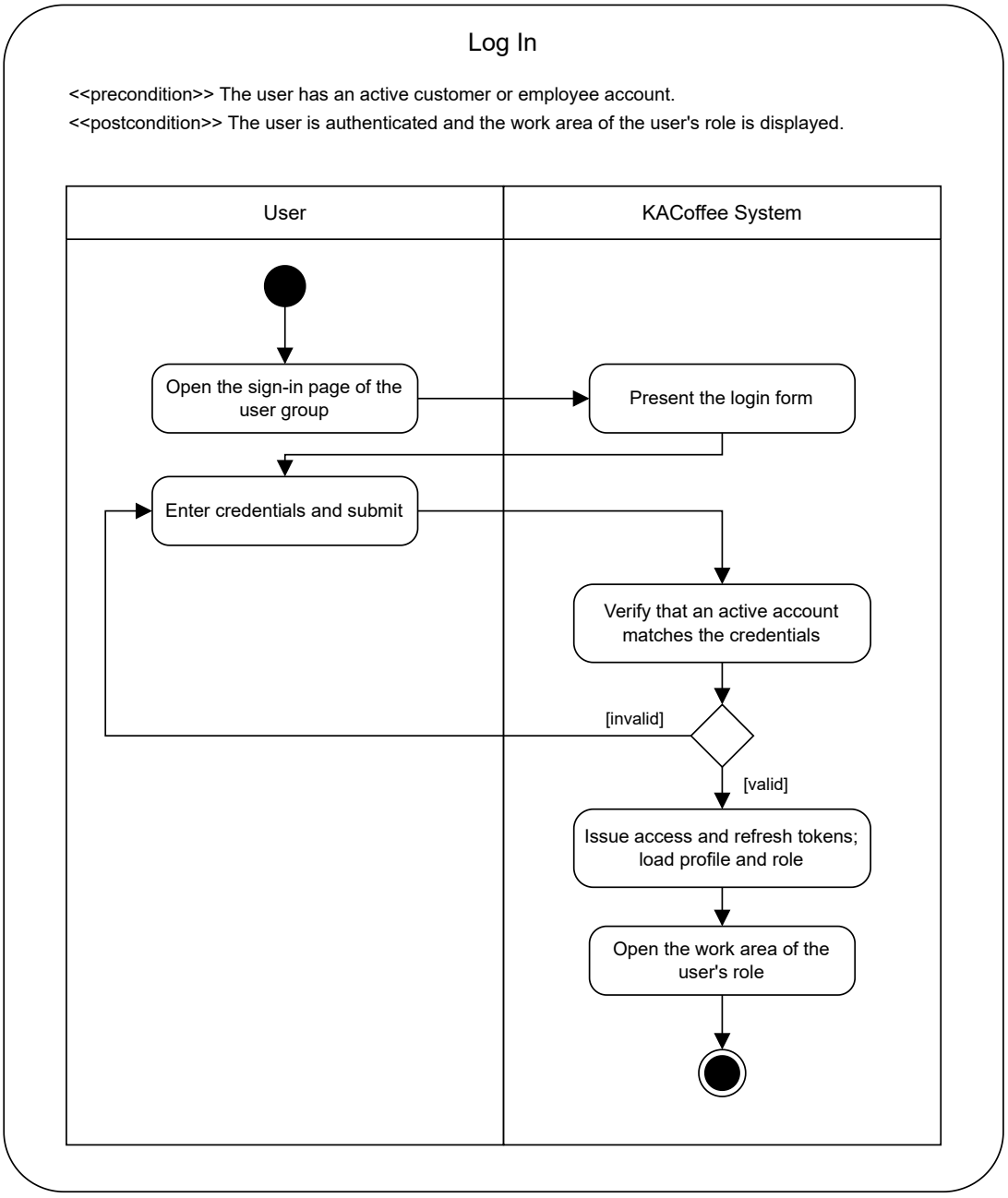


Figure 3: Activity diagram of UC006 Log In

UC007: Log Out

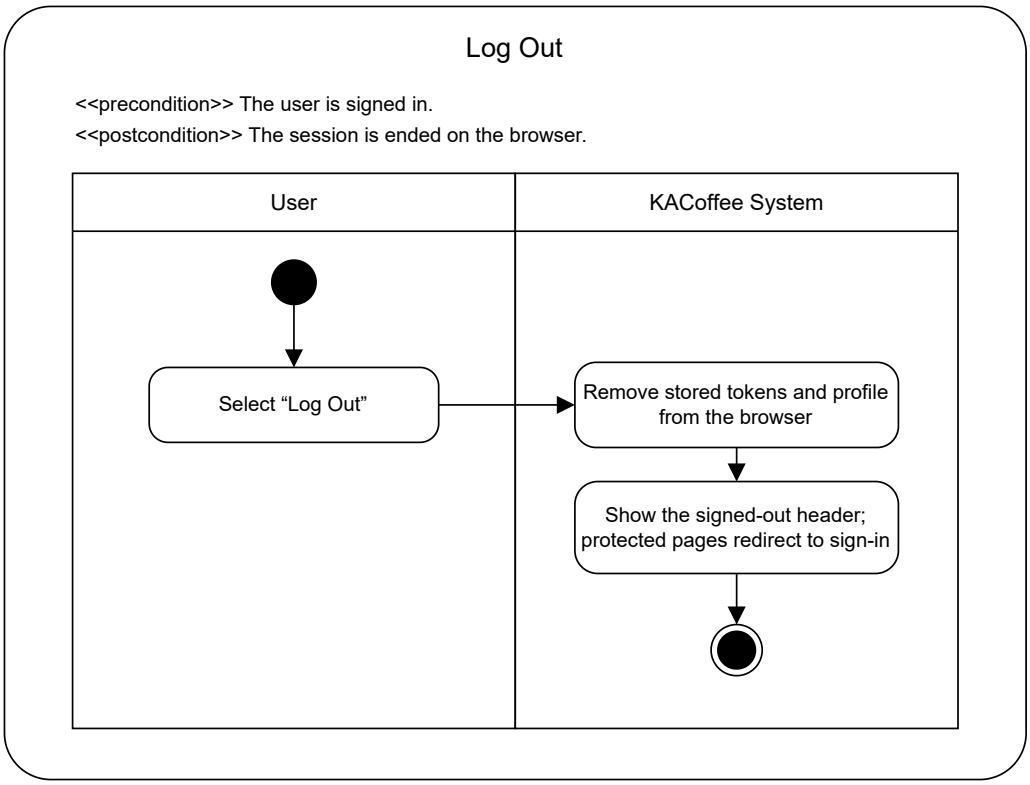


Figure 4: Activity diagram of UC007 Log Out

3.3.2 Customer's Use Cases

UC002: Place Online Order

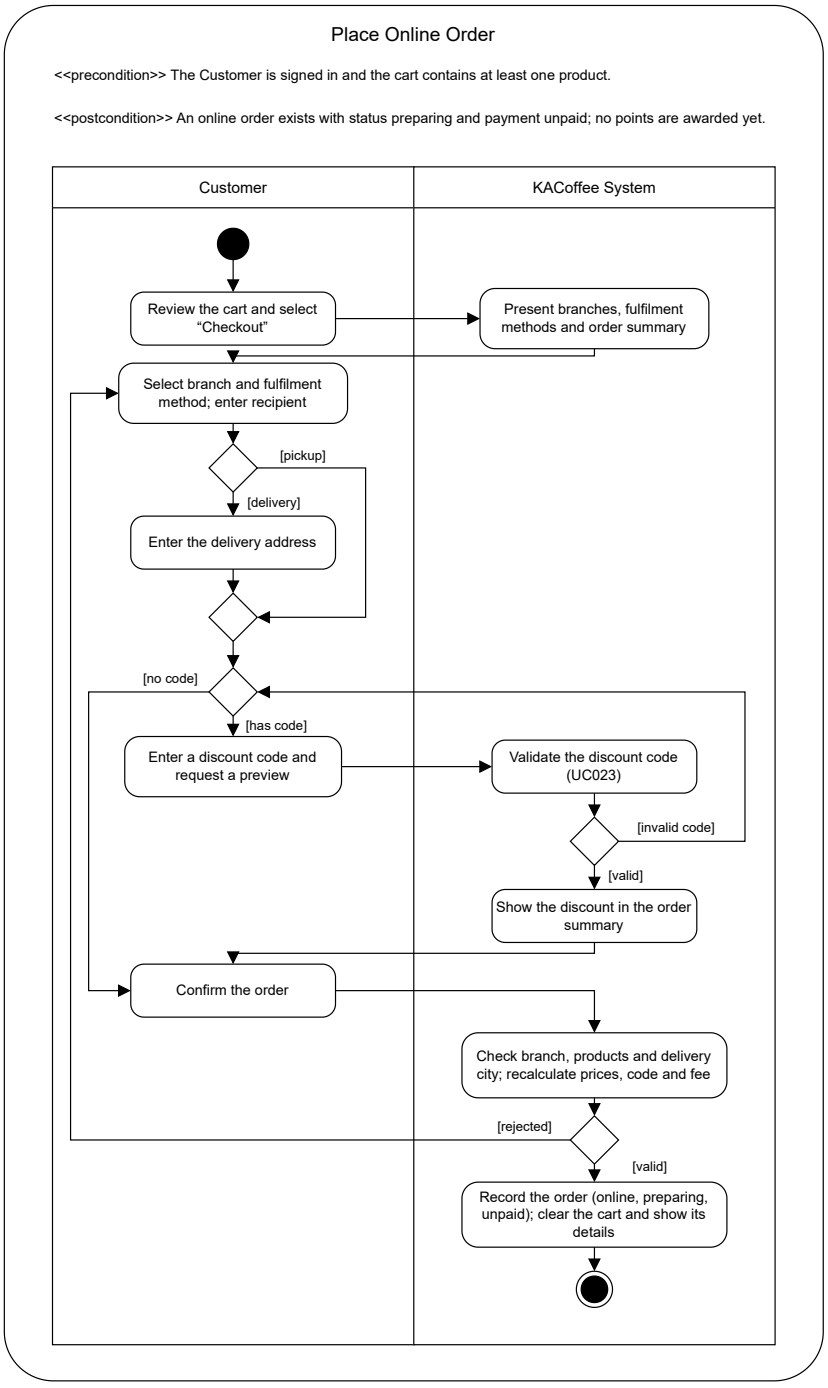


Figure 5: Activity diagram of UC002 Place Online Order

3.3.3 Staff's Use Cases

UC003: Create Staff-Assisted Order

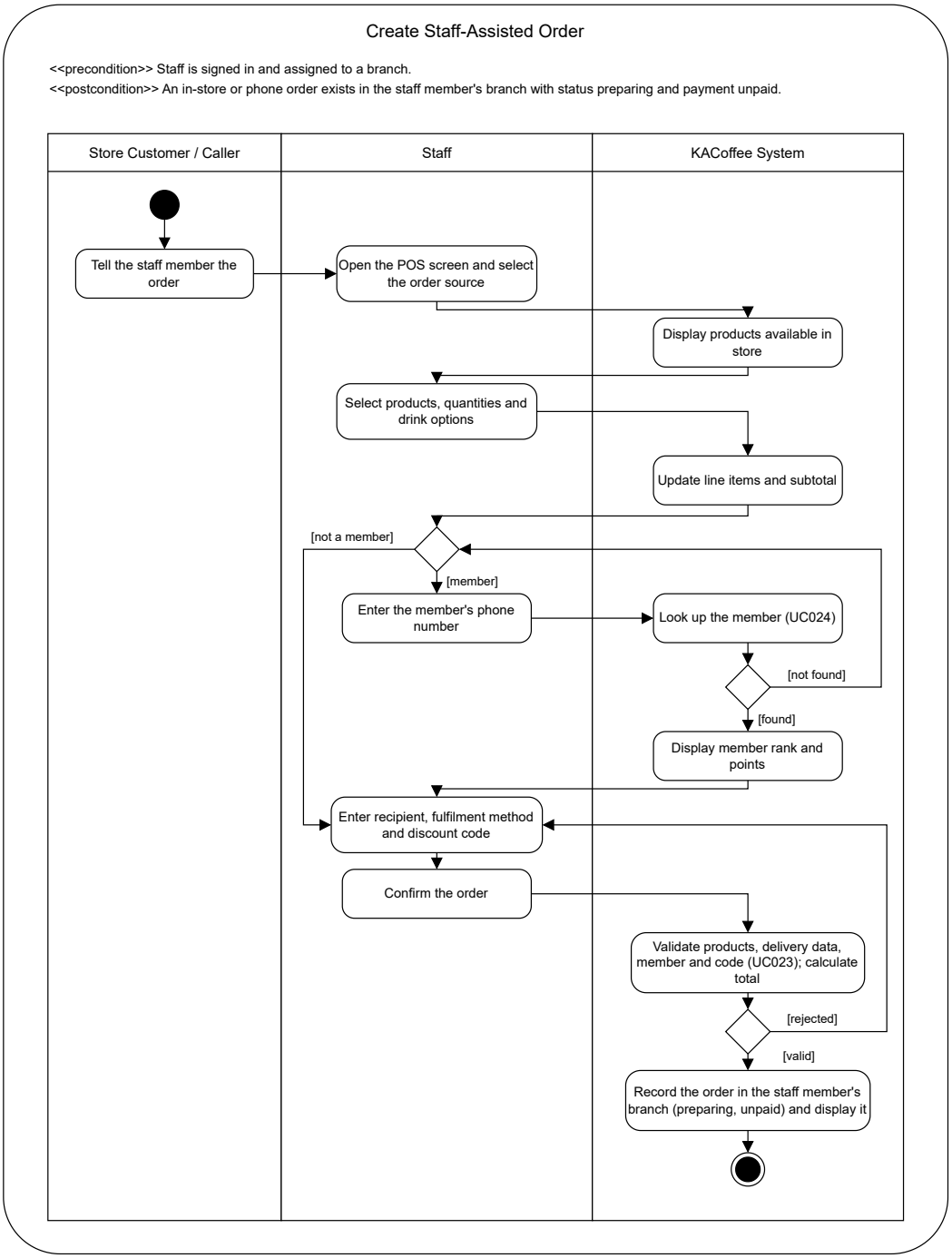


Figure 6: Activity diagram of UC003 Create Staff-Assisted Order

UC008: Update Store Order Status

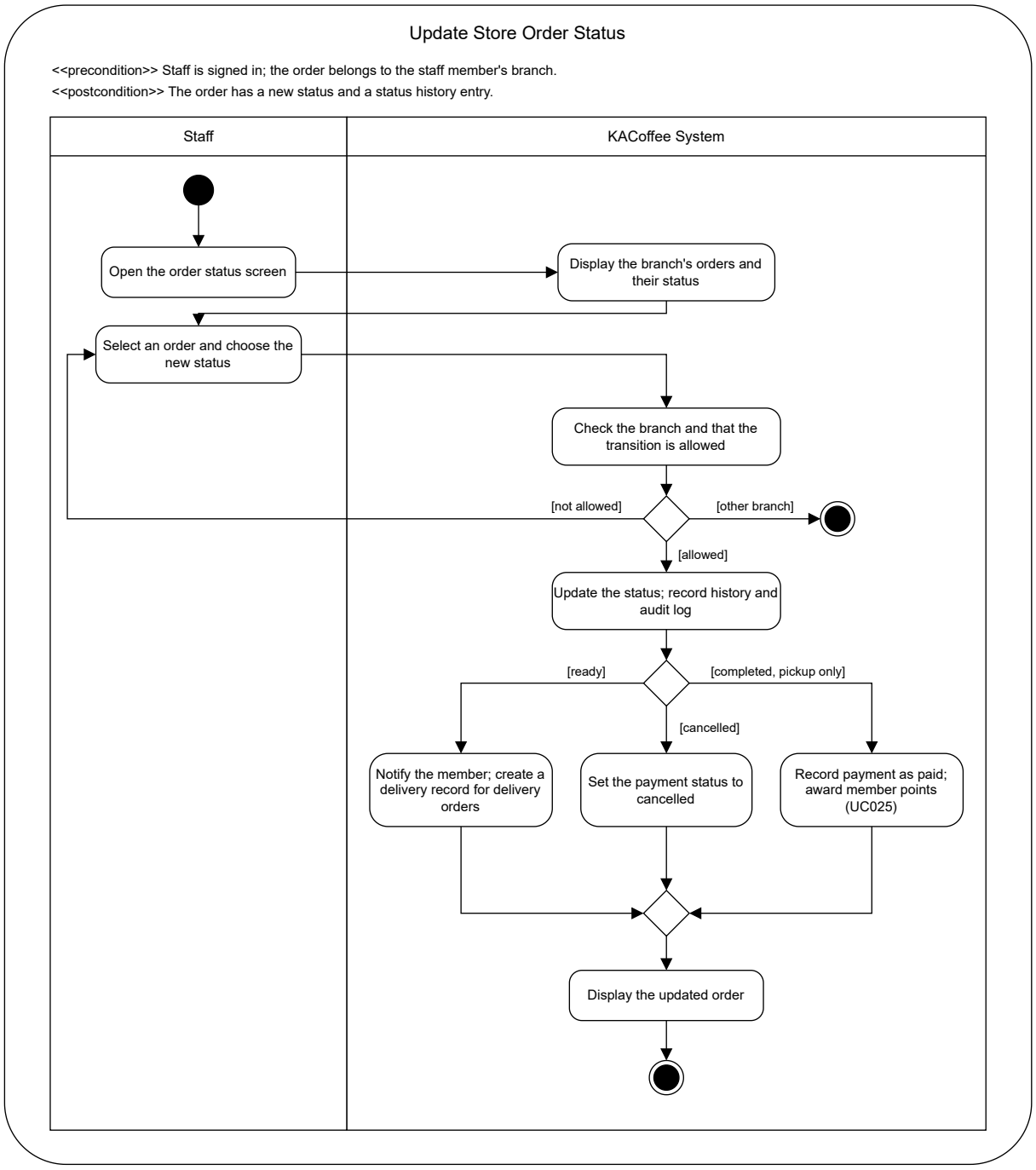


Figure 7: Activity diagram of UC008 Update Store Order Status

UC010: Process Bottle Exchange

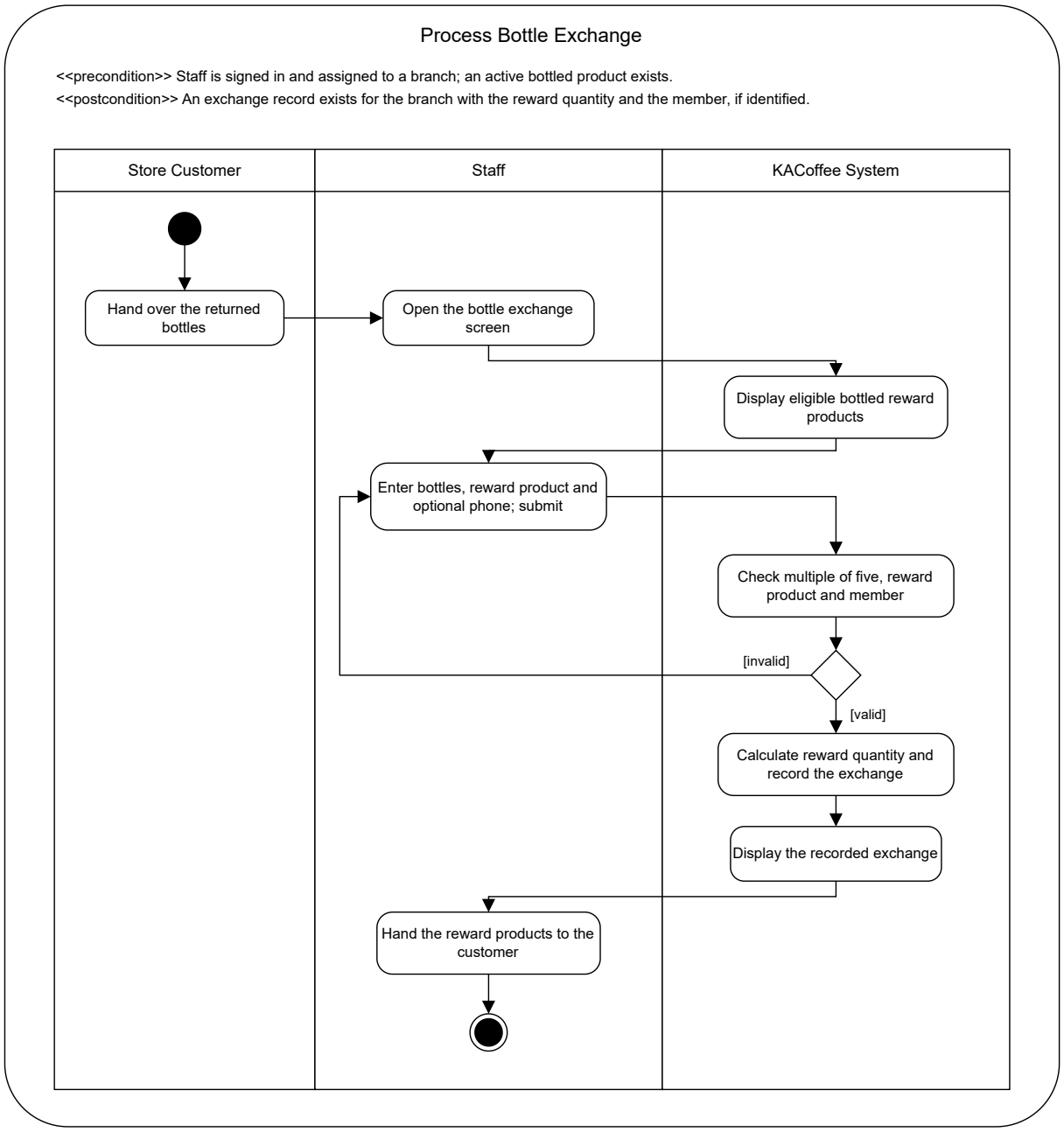


Figure 8: Activity diagram of UC010 Process Bottle Exchange

3.3.4 Shipper's Use Cases

UC005: Update Delivery Status

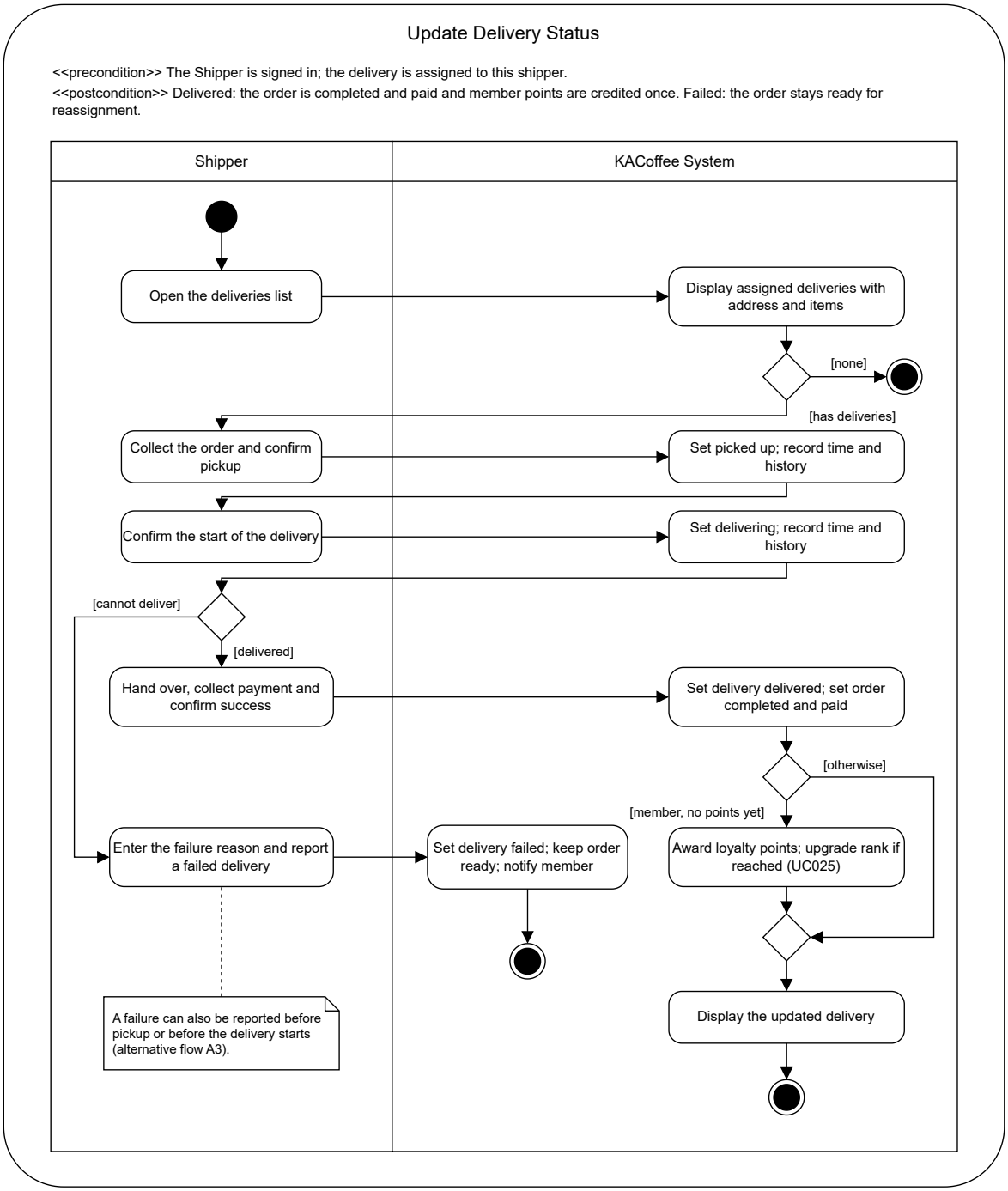


Figure 9: Activity diagram of UC005 Update Delivery Status

3.3.5 Manager's and Admin's Use Cases

UC009: Assign Delivery

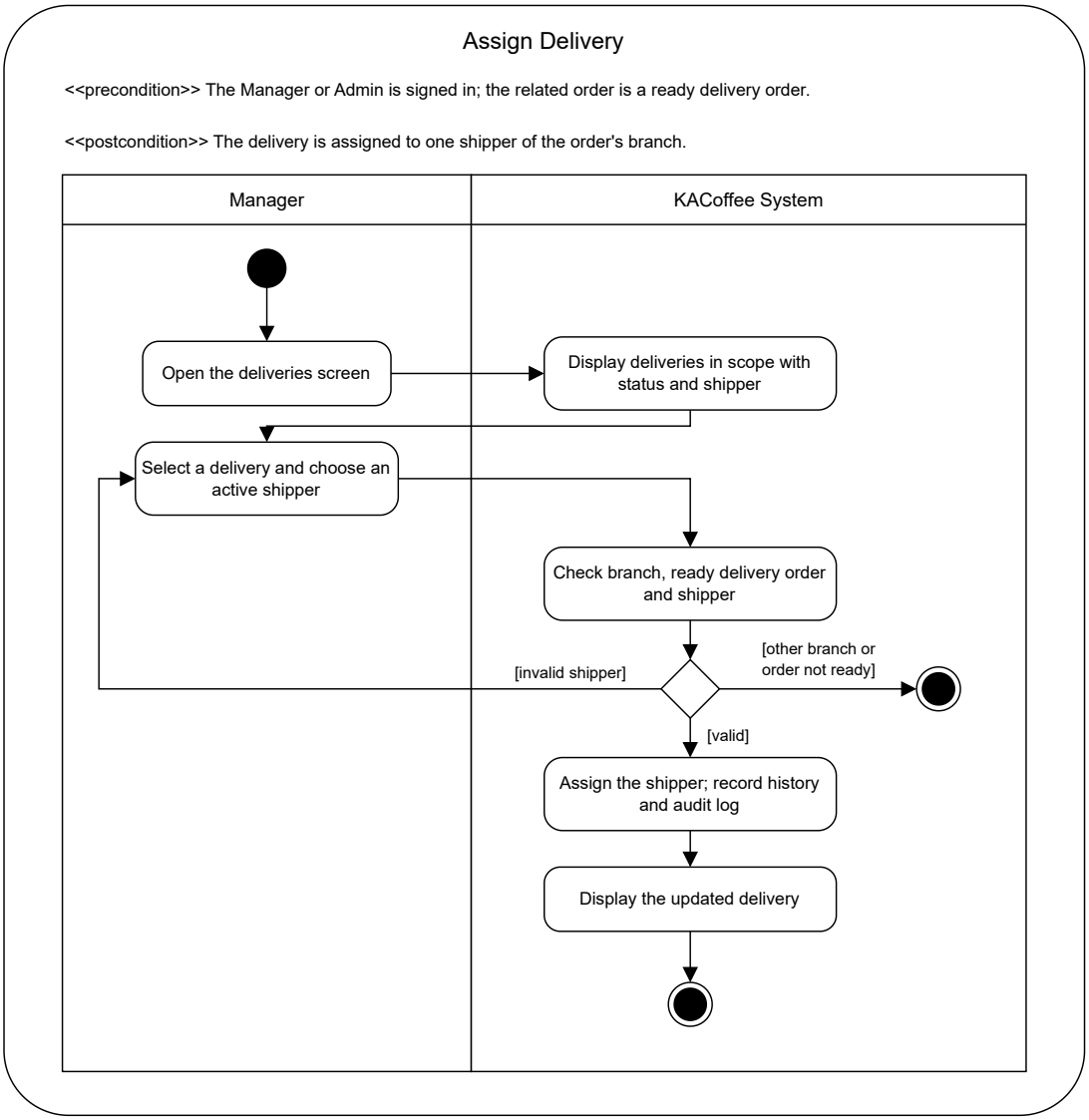


Figure 10: Activity diagram of UC009 Assign Delivery

UC004: Manage Discount Codes

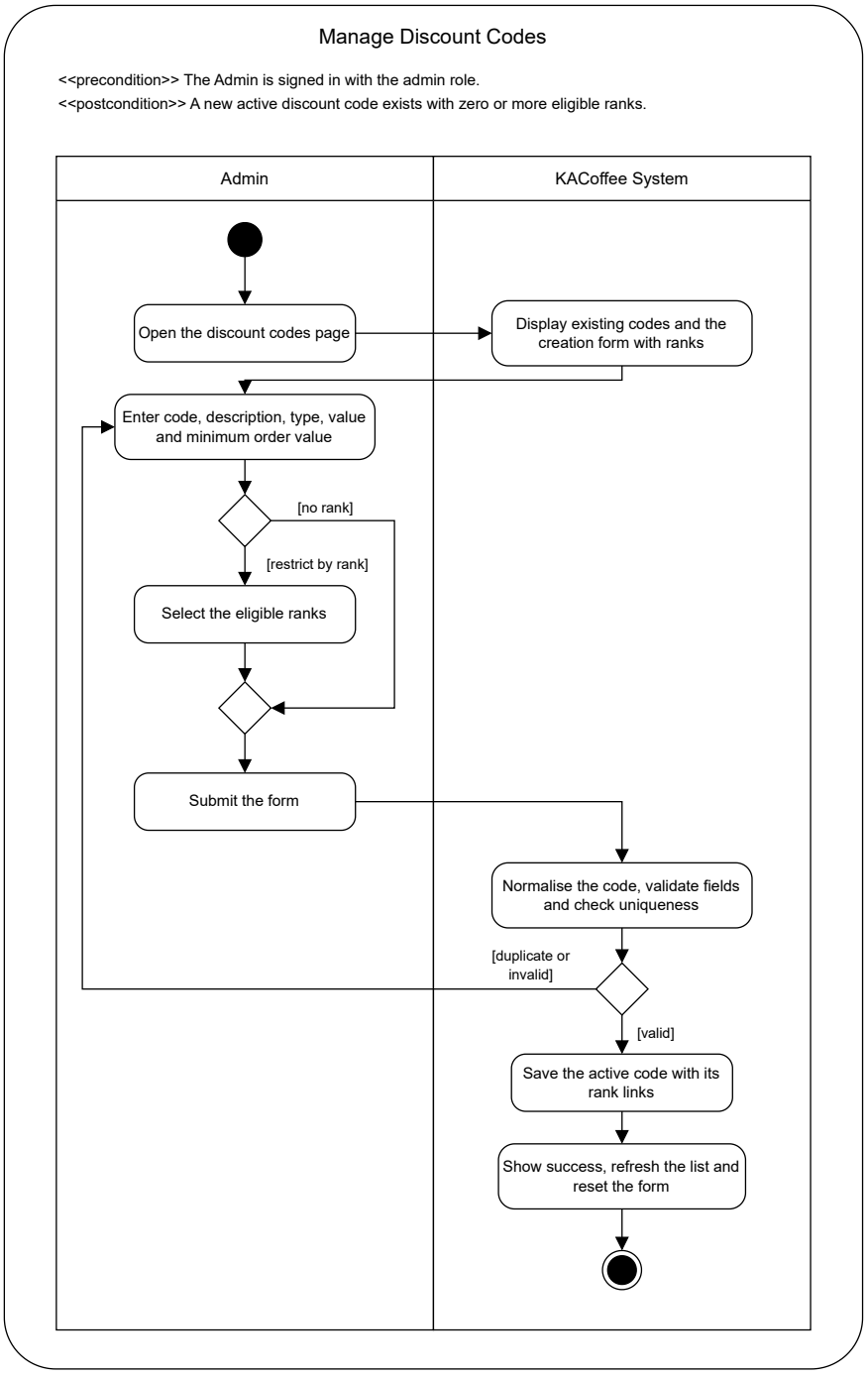


Figure 11: Activity diagram of UC004 Manage Discount Codes

3.4 Sequence Diagram

The sequence diagrams show, for each use case specified in Section 3.2, which objects take part and in which order they exchange messages. They are analysis-level diagrams: the participants are analysis classes rather than concrete source files, and each class carries one of the three stereotypes of the robustness model.

- **Boundary** objects are the screens through which an actor works with the system, such as the checkout page or the POS order screen.
- **Control** objects carry the logic of one use case. Each use case has its own controller, which checks the request, coordinates the entity objects and returns the result to the boundary.
- **Entity** objects hold the data that the system keeps, such as customers, orders, deliveries and discount codes.

The actor only talks to a boundary object, and a boundary object only talks to its controller; entities are reached through the controller. Each diagram is framed and tagged with the code of its use case. Solid arrows with a filled head are calls and dashed arrows are replies, read from top to bottom. Activation bars show when an object is working; the actor stays active for the whole interaction. Combined fragments model the alternative and exception flows of the use case tables: `alt` for mutually exclusive branches, `opt` for an optional step and `loop` for repeated actions. A `ref` fragment points to the diagram of an extending use case, so that discount validation (UC023), member lookup (UC024) and the awarding of loyalty points (UC025) are drawn once and reused. Participants who never use the system directly, such as the store customer in UC003 and UC010, are left out because they exchange no messages with it.

Table 17 lists the analysis classes used in the diagrams.

Table 17: Analysis classes in the sequence diagrams

Stereotype	Classes
Boundary	RegistrationPage, LoginPage, AppHeader, ProtectedPage, CheckoutPage, POSOrderScreen, OrderStatusScreen, BottleExchangeScreen, ShipperDeliveryScreen, DeliveryBoard, DiscountCodePage
Control	RegisterController, LoginController, SessionController, PlaceOrderController, StaffOrderController, OrderStatusController, BottleExchangeController, DeliveryStatusController, AssignDeliveryController, DiscountCodeController, DiscountValidator, MemberLookupController, LoyaltyController

Stereotype	Classes
Entity	Customer, Employee, MembershipRank, RefreshToken, Notification, Branch, Product, Order, Delivery, DiscountCode, PointLedger, BottleExchange, AuditLog

3.4.1 Guest's and Common Use Cases

UC001: Register as Customer

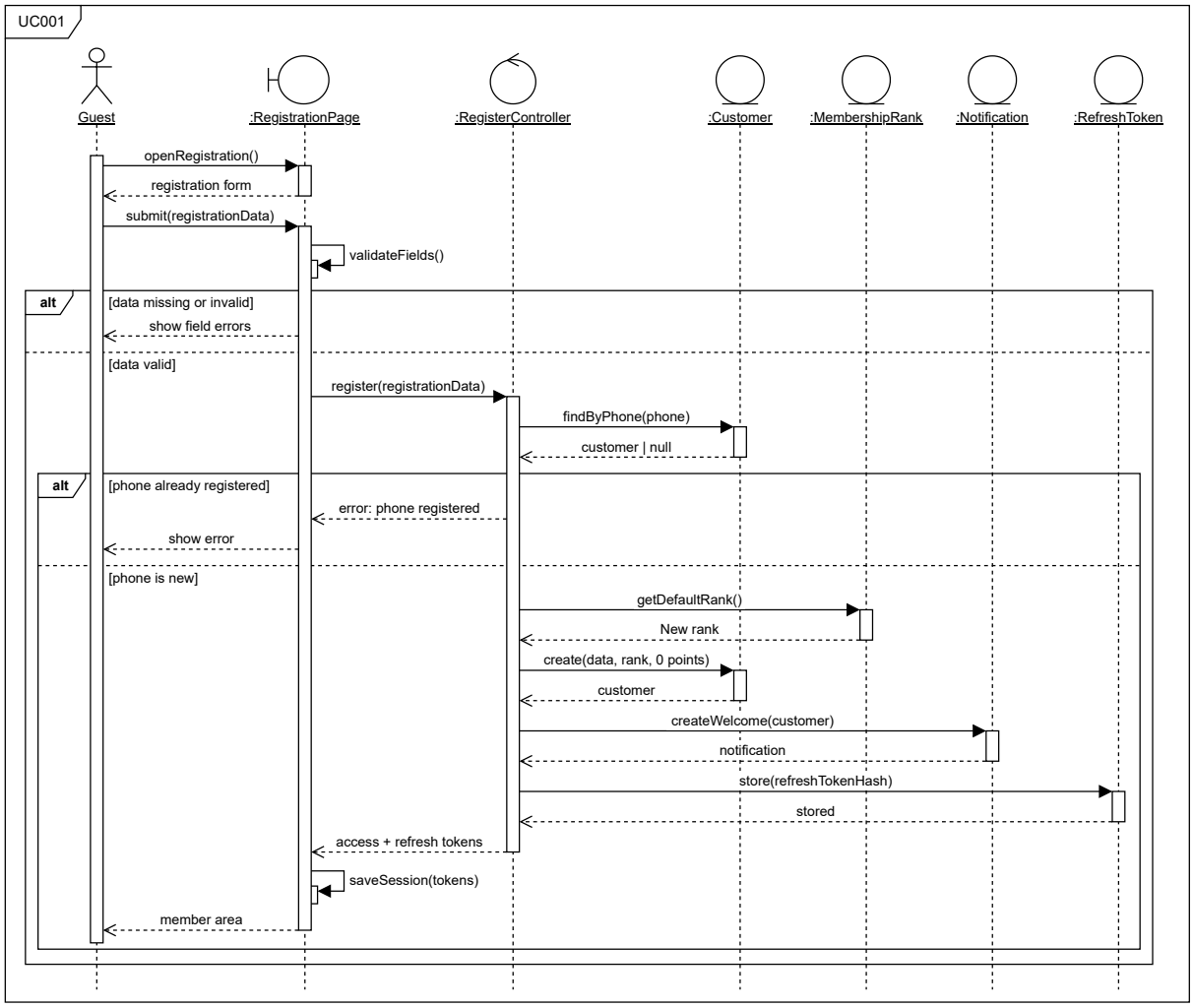


Figure 12: Sequence diagram of UC001 Register as Customer

UC006: Log In

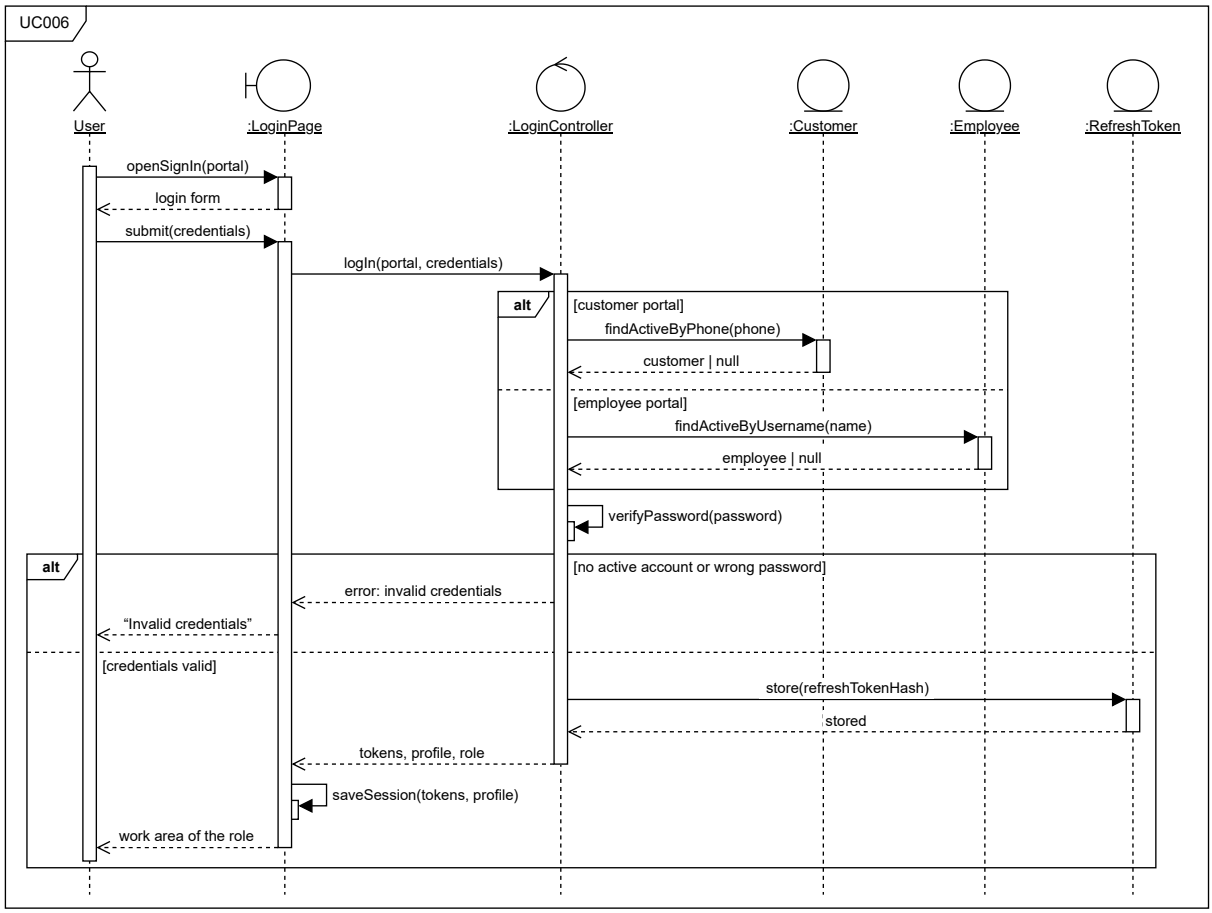


Figure 13: Sequence diagram of UC006 Log In

UC007: Log Out

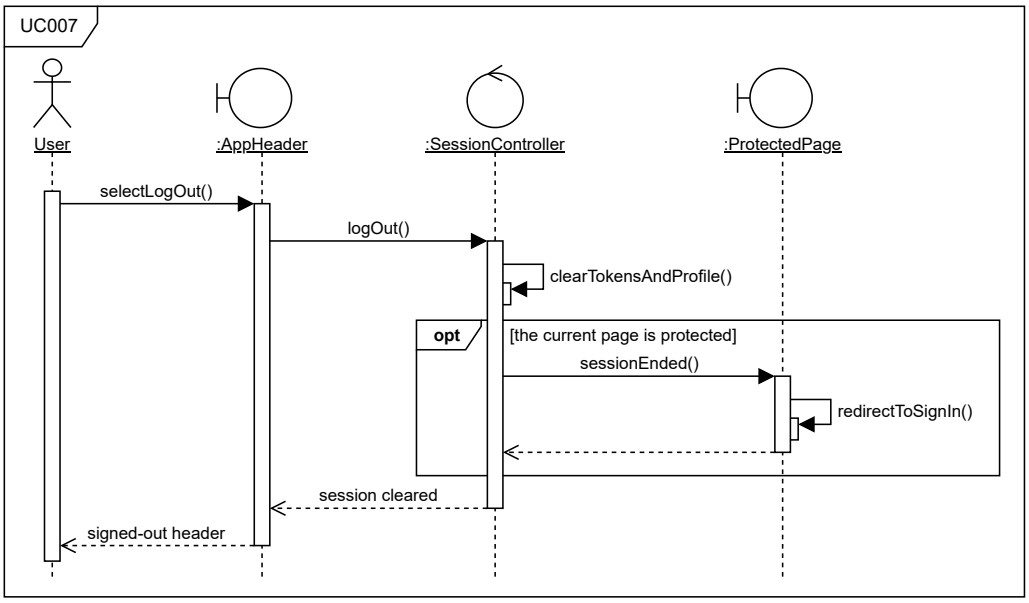


Figure 14: Sequence diagram of UC007 Log Out

3.4.2 Customer's Use Cases

UC002: Place Online Order

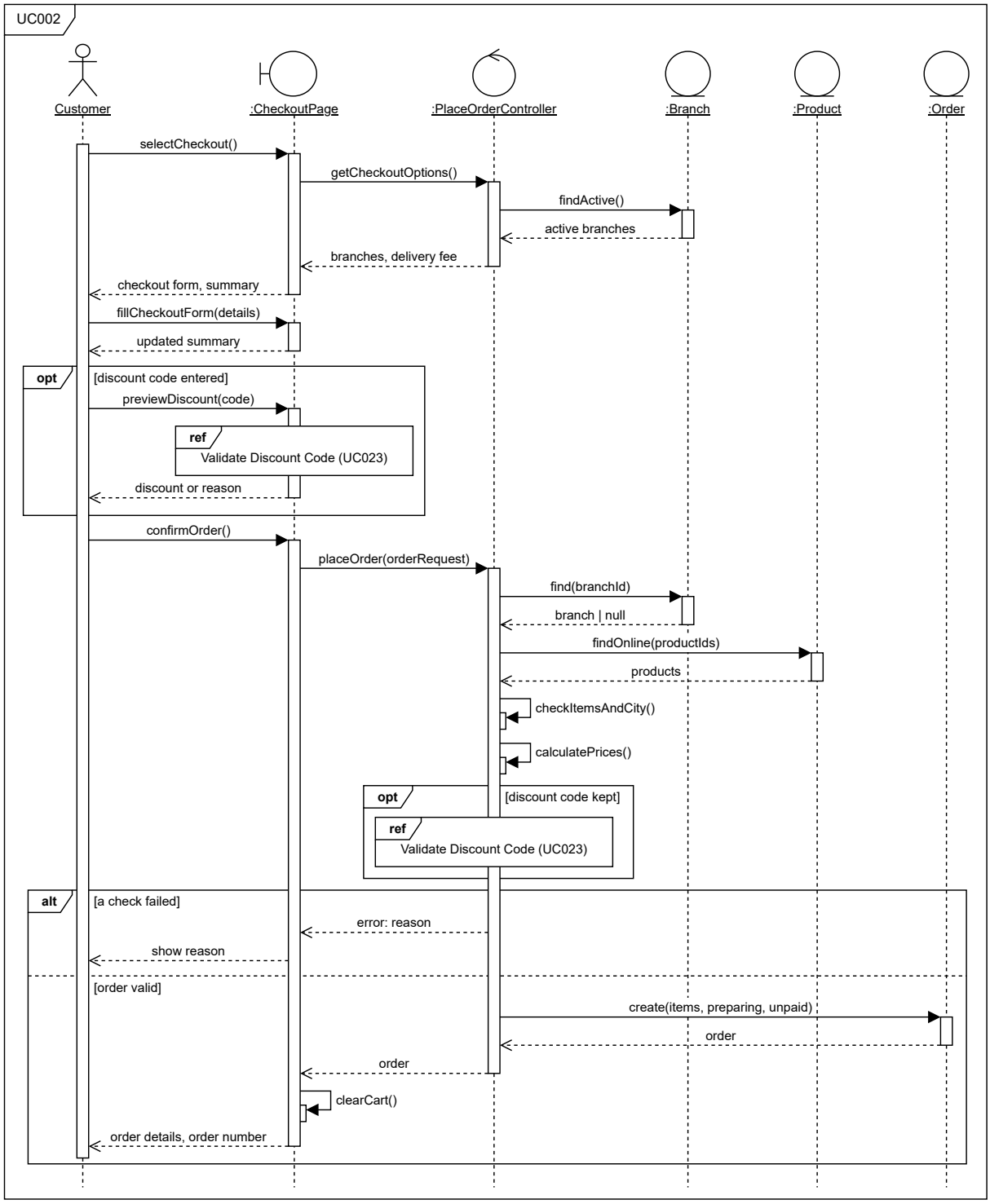


Figure 15: Sequence diagram of UC002 Place Online Order

3.4.3 Staff's Use Cases

UC003: Create Staff-Assisted Order

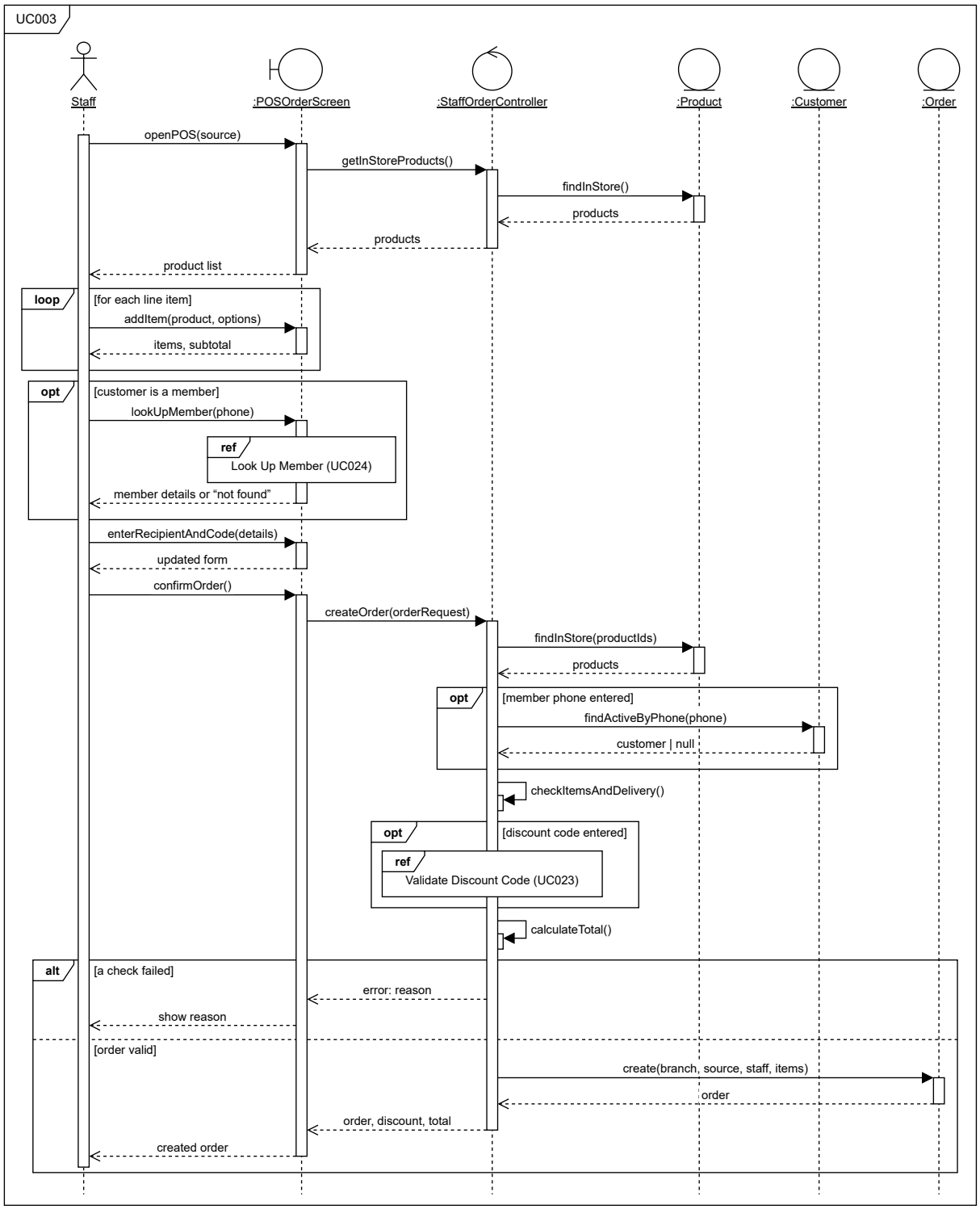


Figure 16: Sequence diagram of UC003 Create Staff-Assisted Order

UC008: Update Store Order Status

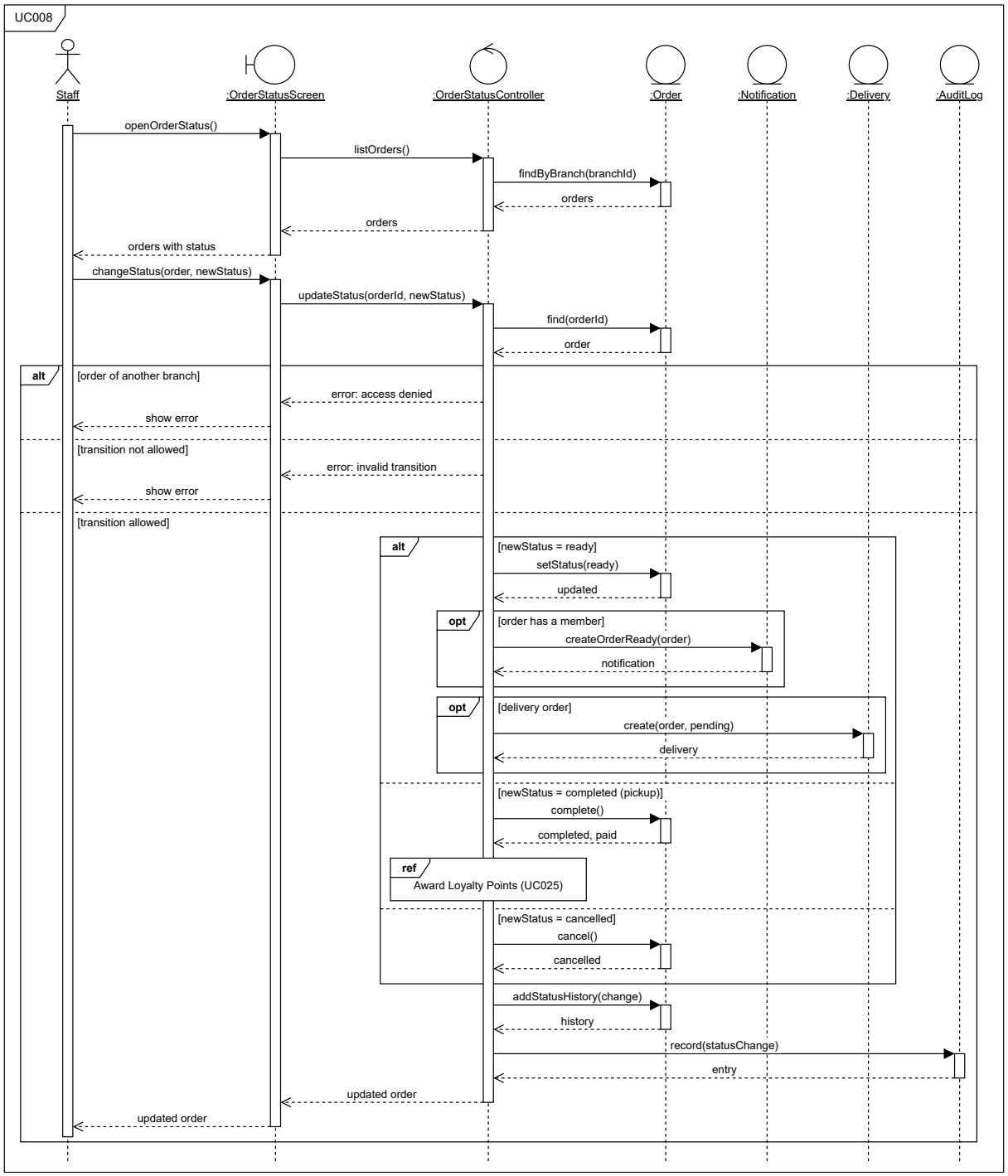


Figure 17: Sequence diagram of UC008 Update Store Order Status

UC010: Process Bottle Exchange

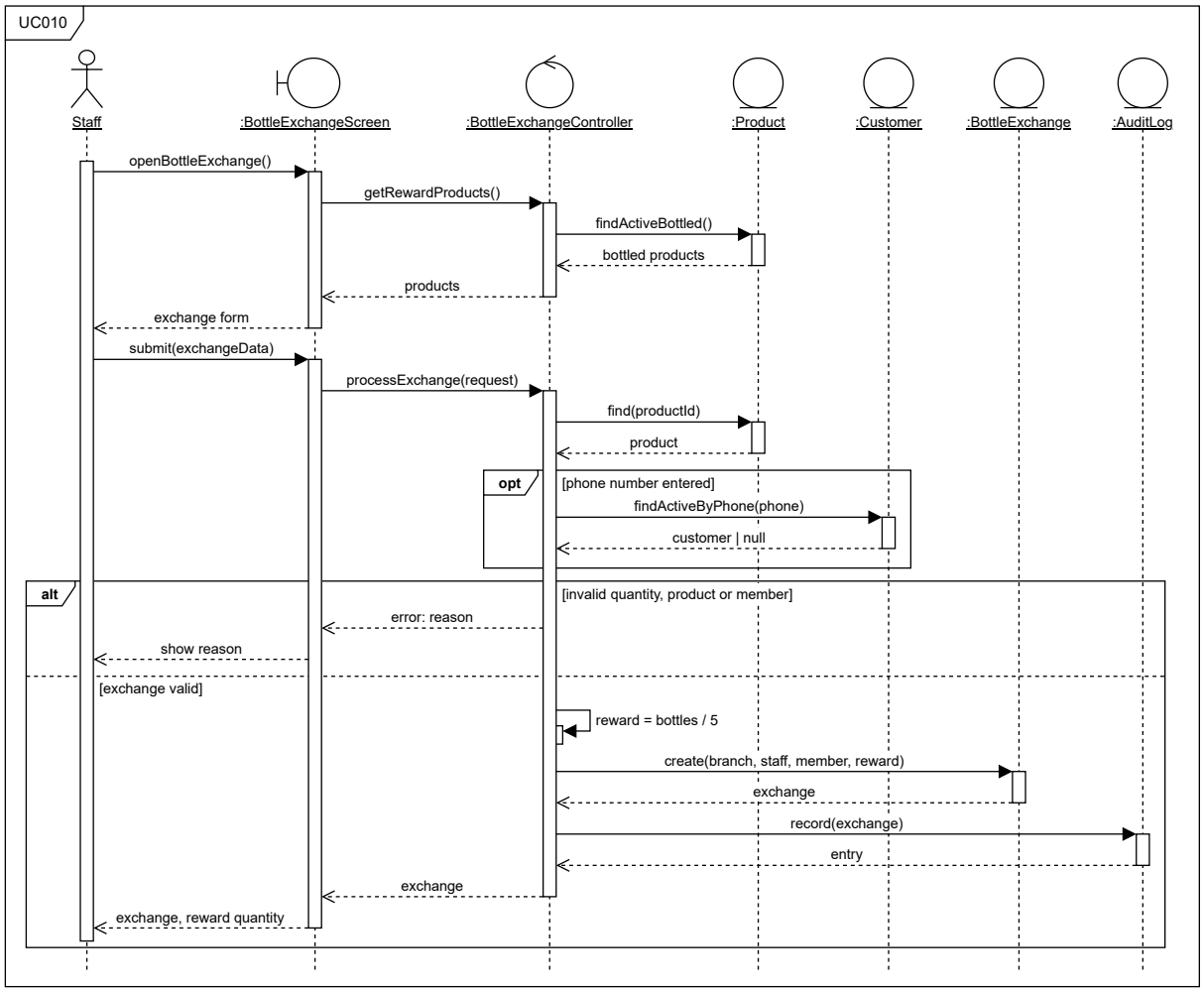


Figure 18: Sequence diagram of UC010 Process Bottle Exchange

3.4.4 Shipper's Use Cases

UC005: Update Delivery Status

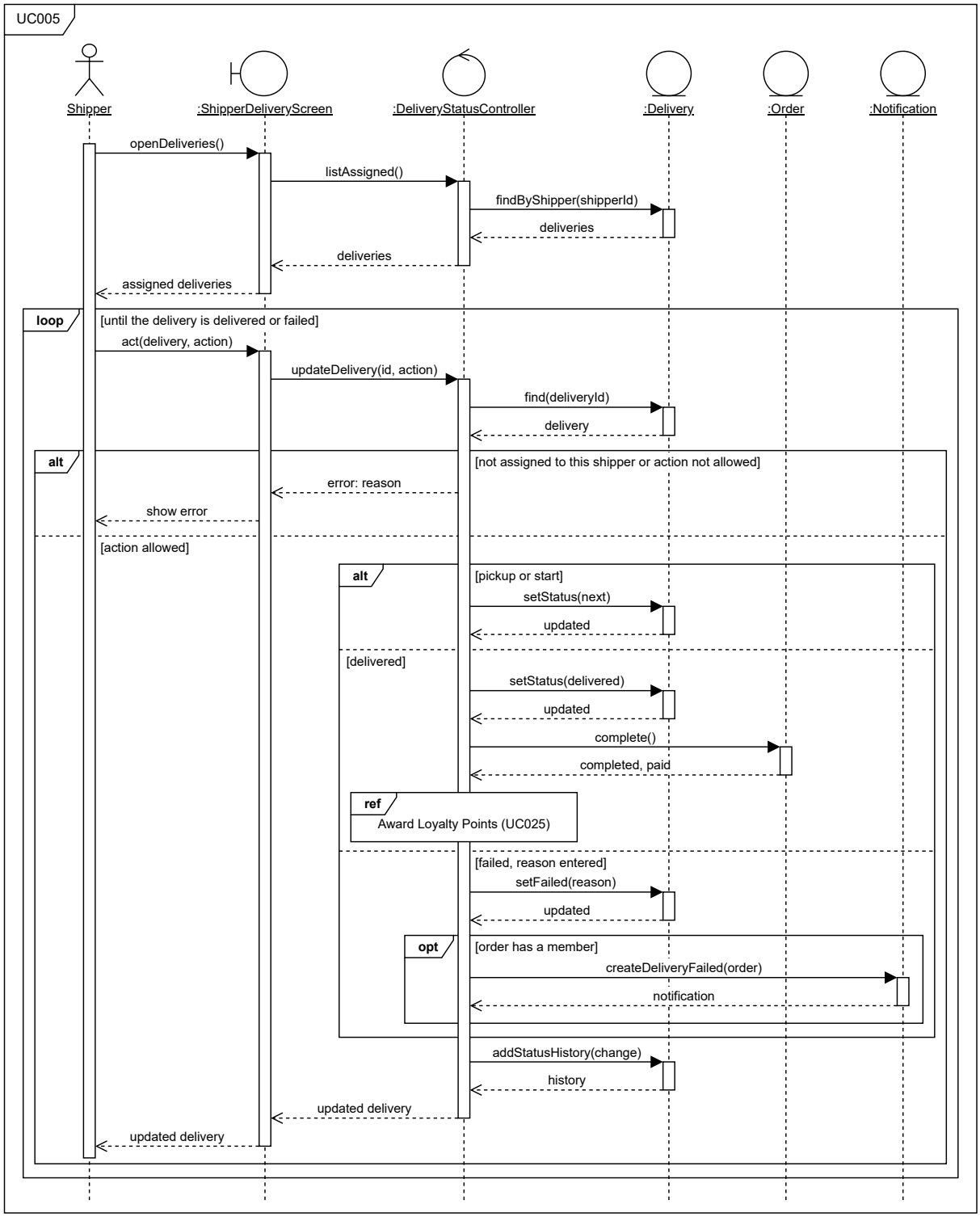


Figure 19: Sequence diagram of UC005 Update Delivery Status

3.4.5 Manager's and Admin's Use Cases

UC009: Assign Delivery

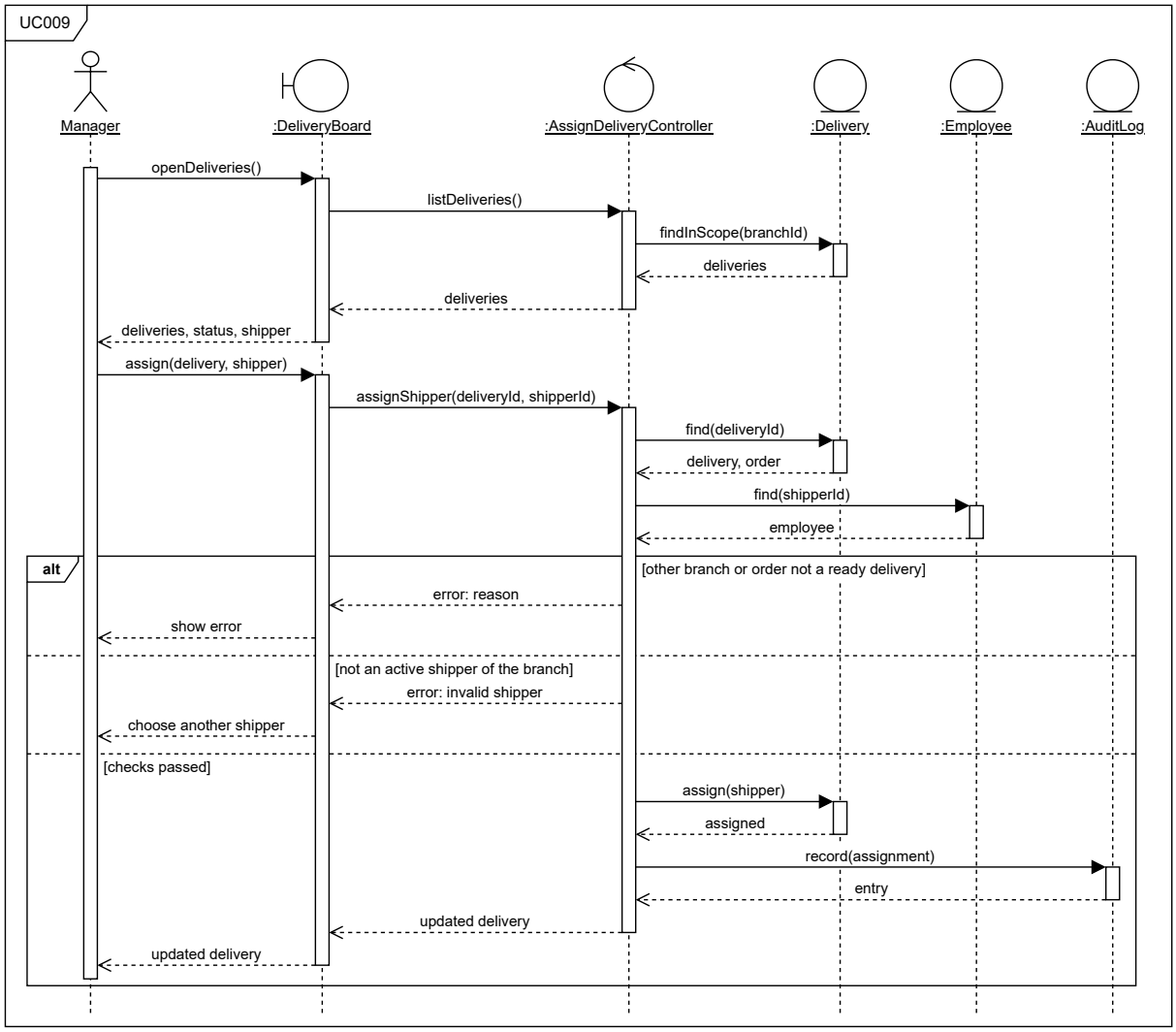


Figure 20: Sequence diagram of UC009 Assign Delivery

UC004: Manage Discount Codes

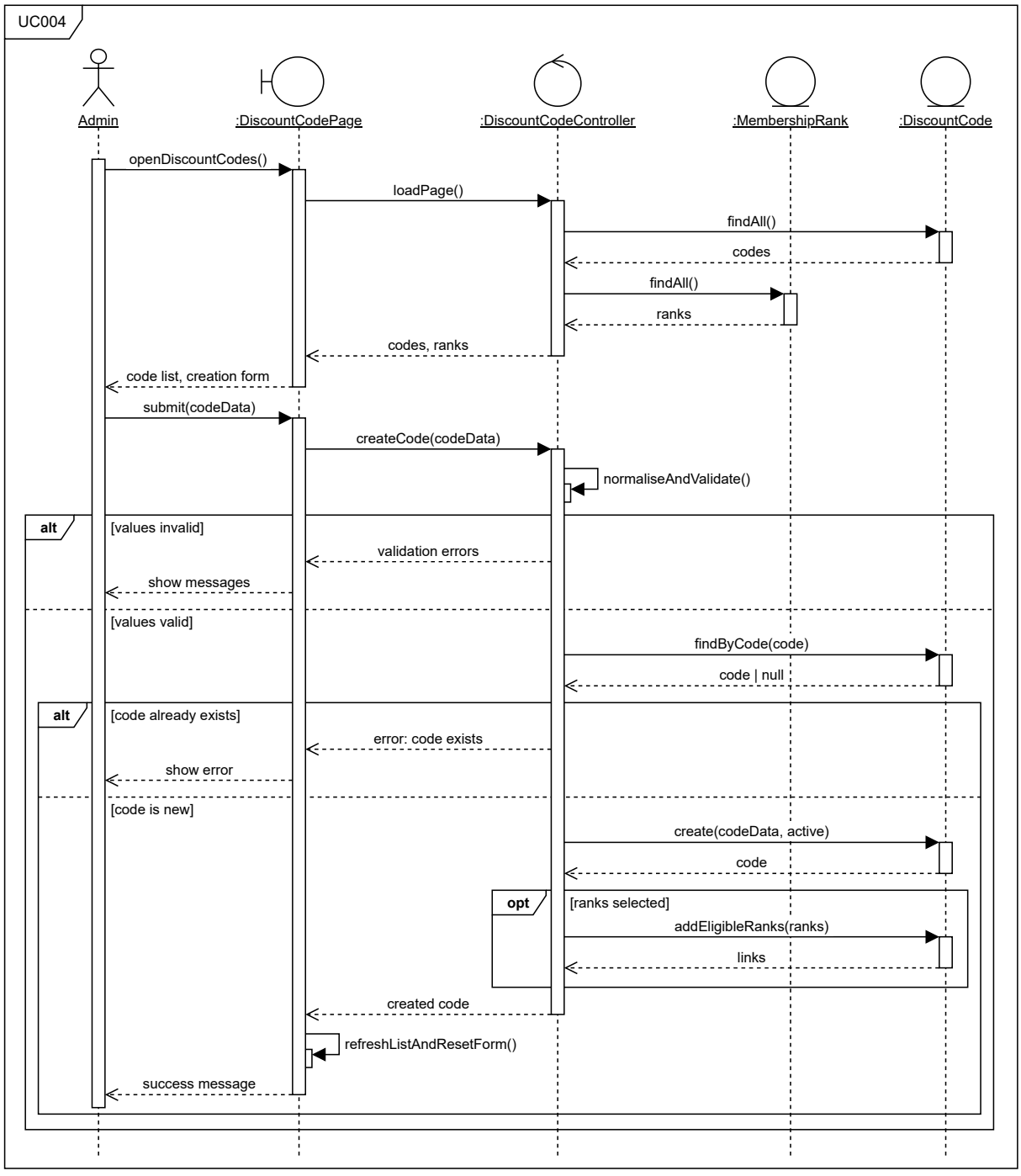


Figure 21: Sequence diagram of UC004 Manage Discount Codes

3.4.6 Shared Extension Use Cases

These three diagrams are referenced by the `ref` fragments above. The incoming message enters through a gate on the frame, because the caller is the controller of the base use case.

UC023: Validate Discount Code

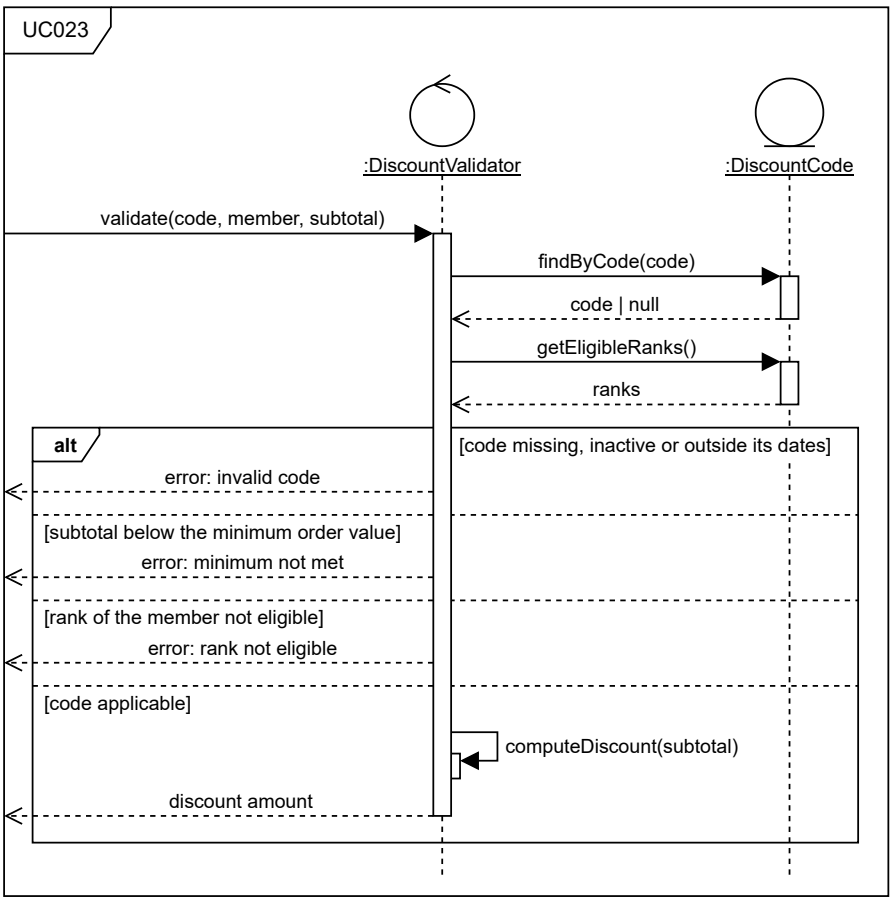


Figure 22: Sequence diagram of UC023 Validate Discount Code

UC024: Look Up Member

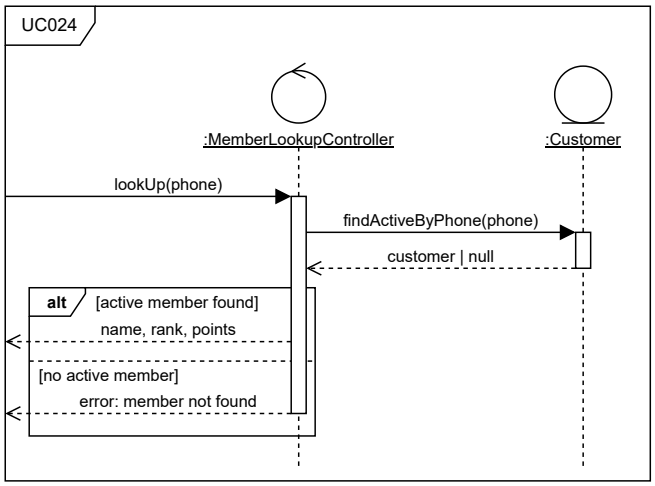


Figure 23: Sequence diagram of UC024 Look Up Member

UC025: Award Loyalty Points

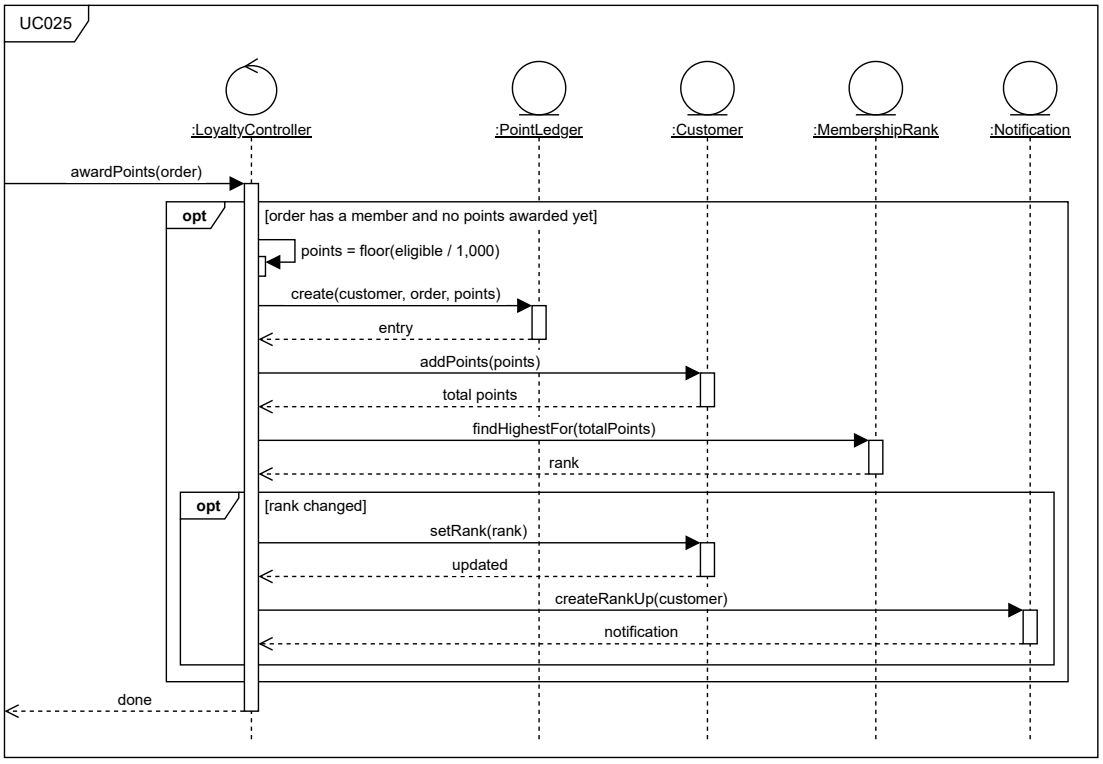


Figure 24: Sequence diagram of UC025 Award Loyalty Points